When good balance goes bad: A discussion of common pitfalls when using entropy balancing

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Abstract

For many accounting research questions, empirical researchers cannot randomly assign observations to treatment conditions or identify a quasi-experimental setting. In these cases, entropy balancing (Hainmueller 2012) is an increasingly popular statistical method for identifying a control sample that is nearly identical to the treated sample with respect to observable covariates. In this paper, we compare entropy balancing's approach of reweighting control sample observations to ordinary least squares and propensity score matching. We demonstrate that researchers applying entropy balancing in empirical settings involving panel data with features common in accounting research may encounter implementation issues that render the resulting estimates sensitive to relatively minor changes in the control sample or the research design. Using the setting of estimating the Big-N audit fee premium, we empirically demonstrate these issues and propose solutions.

Keywords: *entropy balancing; matching; causal inference; covariate balance; audit fees*

JEL: C18, M4

Data availability: Data are available from public sources cited in the text.

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I. INTRODUCTION

A recent trend in empirical-archival accounting research emphasizes drawing causal inferences in a range of research settings.[1] Despite this trend, empirical accounting researchers are frequently unable to randomly assign observations to treatment conditions or identify a quasi-experimental setting for a given research question, resulting in a need to identify samples of observations that appear as similar as possible with respect to underlying covariates. Following its introduction by Hainmueller (2012), entropy balancing (EB) has become one of the most widely used methods for addressing covariate imbalance across a treatment and control sample of interest, with Google Scholar showing nearly 1,800 studies across disciplines, and 128 studies in accounting specifically, that cite Hainmueller (2012).[2] Like other matching methods, such as propensity score matching (PSM, Rosenbaum and Rubin 1983), the ultimate goal of EB is to identify a control sample that appears similar to a treatment sample of interest. However, instead of estimating a distance metric (e.g., a difference in propensity scores), EB estimates non-negative weights for each observation in the control sample such that the specified moments (e.g., mean, variance, and/or skewness) of the covariate distributions of the weighted control sample are nearly identical to those of the treated sample.

Given its ease of implementation and ability to ensure nearly exact covariate balance without the need to specify a propensity score model or identify a (potentially noisy) one-to-one match for each treated observation, EB's popularity is understandable. However, it is not always straightforward to apply EB "as is" using panel data, especially for research questions where the

[1] Gow, Larcker, and Reiss (2016) survey accounting research that draws causal inferences.
[2] Google Scholar listed the following references on August 1, 2021 (by journal): 32 in *The Accounting Review*, 12 in the *Journal of Accounting and Economics*, 3 in the *Journal of Accounting Research*, 7 in the *Review of Accounting Studies*, 21 in *Contemporary Accounting Research*, and 53 in other journals. We also find 156 cites to McMullin and Schonberger (2020), which focuses on implementing entropy balancing in accounting settings, as of this date.

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assignment to treatment is poorly understood, where there is little overlap in the distributions of key covariates across treatment and control samples, where the treatment is continuous, or where the construct of interest is the interaction between two independent variables. In each of these cases, EB displays drawbacks without obvious solutions for empirical researchers.

The goal of this study is to highlight common misconceptions in the application of EB by first comparing EB with common alternative approaches to address covariate imbalance, with a focus on ordinary least squares (OLS) regression and PSM.[3] We then discuss issues with how EB assigns weights to observations in settings with substantial distributional differences between treatment and control samples and in settings with panel data, which are common features in empirical accounting studies. In these settings, while EB automatically generates observational weights that balance treatment and control samples (by construction), we show that examining the assigned weights is important. Two common misunderstandings regarding these weights are: (1) EB assigns a meaningful weight that is strictly greater than zero to all observations (i.e., all observations are "used" in estimating a treatment effect), and (2) researchers should use the weights without considering their magnitude. On this latter point, we conclude that in some cases, researchers will likely prefer to discard observations with extreme weights.

Because our study focuses on issues when using EB to address covariate imbalance in observational studies, our paper is similar to recent work that addresses the implementation of PSM (Shipman, Swanquist, and Whited 2017) and EB (McMullin and Schonberger 2020) in empirical accounting settings. In contrast to these studies, our focus is on providing practical

[3] We use PSM as a fundamental matched-sample benchmark in our tests because it is relatively easy to understand, is commonly used, and adopts a more discrete form of matching relative to entropy balancing. Further, simulation analyses by Morgan and Winship (2015, Table 5.6) show that PSM performs better than alternative matching approaches, such as coarsened exact match (CEM). That said, we do not directly compare EB to related forms of sample reweighting that rely on more continuous versions of PSM-derived weights, such as the covariate balancing propensity score (Imai and Ratkovic 2014). We thank an anonymous referee for making these points.

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implementation guidance for EB along with a detailed empirical example (with corresponding Stata program code) designed to directly compare alternative weighting approaches available to empirical researchers. As a result, we hope the study is useful both for students in research-oriented doctoral programs as well as researchers new to EB as a tool.

To demonstrate common implementation issues and proposed solutions empirically, we use the setting of estimating the Big-N audit fee premium, where the treatment sample of interest comprises Big-N auditor client firms and the control sample is all firms receiving audits from a non-Big-N auditor. As a continuous measure of audit quality (or audit effort), we focus our analyses on audit fees as the outcome of interest with fundamental determinants (i.e., covariates) of audit fees drawn from DeFond and Zhang's (2014) survey of the audit literature. In addition to serving as an accessible example for most researchers, the Big-N setting allows us to focus on the implementation of EB using panel data. Of the 128 accounting studies citing Hainmueller (2012), we find that 47 (37 percent) mention a difference-in-differences estimate, consistent with empirical accounting researchers applying EB in panel settings that are largely unaddressed by both Hainmueller (2012) and McMullin and Schonberger (2020).

The remainder of the paper is organized as follows. In Section 2, we describe the Rubin Causal Model (RCM) as a framework for drawing causal inferences and situate EB as a sample reweighting tool within this framework. We then summarize the key features of EB as a method to ensure covariate balance across a treatment and control sample of interest, including an example using simulated data to illustrate the weighting process used by EB relative to OLS and PSM approaches. This latter portion of Section 2 relies on Hainmueller (2012) and McMullin and Schonberger (2020) as part of the overview of EB, before providing a practical discussion for researchers and students seeking to apply EB in an empirical research setting of interest.

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Section 3 describes the data and sample for the Big-N audit fee setting used for our analysis along with audit fee models drawn from DeFond and Zhang (2014). In Section 4, we present summary statistics for covariate imbalance and baseline estimates of the Big-N audit fee premium for OLS, PSM, and EB models.

In Section 5, we use the baseline models described and estimated in Sections 3 and 4, respectively, to describe two common issues when implementing EB. First, we rely on pooled sample estimates to illustrate an issue whereby EB assigns extremely large weights to a handful of control sample observations in order to achieve covariate balance.[4] We discuss the features of the underlying dataset that result in the assignment of extreme weights before exploring three solutions designed to reduce the influence of extreme weights across a range of research settings. Second, we describe issues when applying EB in a panel data setting. In particular, we focus on how EB assigns weights across periods in order to achieve covariate balance and discuss the benefits and drawbacks of assigning weights in individual periods of the panel.

Section 6 summarizes the advantages and disadvantages of EB and provides our key implementation recommendations for researchers using EB as a sample reweighting tool. Briefly, we recommend that when using EB, researchers should ensure covariate balance in individual years of a panel dataset when constructing difference-in-differences estimates and should examine summary statistics for the distribution of observational weights to determine whether extreme weights are present in a given setting. To address issues with extreme weights, researchers should either restrict the treatment and control samples to a given section of the covariate distribution (to construct an estimate of a local average treatment effect) or consider

[4] The issue of overweighting is not unique to EB. In other approaches where researchers use one-to-one matching with replacement (e.g., PSM or attributes-based matching), the same control observation can be matched multiple times to differing treatment observations, resulting in large observational weights.

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down-weighting extreme observations in the treatment sample of interest, either by using EB to assign weights to the treatment sample (rather than the control sample) or by trimming observations from the treatment sample that lack a valid counterfactual.[5]

II. BRIEF CONCEPTUAL DISCUSSION OF ENTROPY BALANCING

Rubin Causal Model

The Rubin Casual Model (RCM) defines a unit-level causal effect τ_i as the difference between two potential outcomes: the outcome if a unit receives a treatment of interest $y_i(T = 1)$ and the alternative outcome if the same unit does not receive the treatment $y_i(T = 0)$.[6] That is,

$$\tau_i = y_i(T = 1) - y_i(T = 0). \quad (1)$$

The fundamental problem of causal inference is that as researchers, we observe only one potential outcome for each unit of observation (at most). As we cannot observe both potential outcomes, estimating a causal effect requires examining multiple units.

The RCM identifies two primary assumptions that researchers make when using multiple units to estimate the missing potential outcomes required to identify a causal effect: the stable unit treatment value assumption (SUTVA) and the strongly ignorable treatment assignment (SITA) assumption. Imbens and Rubin (2015, p. 10) state SUTVA as: "The potential outcomes for any unit do not vary with the treatments assigned to other units, and for each unit, there are no different forms or versions of each treatment level which lead to different potential outcomes." The SITA assumption underlying the RCM (which is the primary focus of reweighting techniques, such as matching) relates to the mechanism controlling the assignment to treatment. Imbens and Rubin (2015) note that under this assumption the assignment

[5] Trimming observations as part of the matching program is also referred to as matching within common support.
[6] Although these concepts were first proposed by Neyman (1923) in the context of randomized experiments, Rubin extended them to a general framework applicable to both experiments and observational (archival) studies. For helpful introductions to the RCM, see Holland (1986), Morgan and Winship (2015), and Imbens and Rubin (2015).

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mechanism is: individualistic, probabilistic, and unconfounded. The individualistic assumption holds that the probability of a unit being treated is not dependent on the treatment or outcome of other units. The probabilistic assumption holds that each unit has a probability of being in each treatment condition that is strictly between zero and one, such that probabilistic assignment is not met if a firm receives the treatment of interest (e.g., a Big-N audit) with probability zero or one. In these latter cases, only one potential outcome exists and a unit-level causal effect is undefined. The unconfoundedness assumption holds that the assignment to treatment does not depend on the potential outcomes (e.g., each firm does not base its choice of auditor on the potential level of audit quality under each condition). Importantly, this portion of the SITA assumption is at odds with the idea that firms optimize as economic agents.

For randomized control trials, the assignment to treatment should meet SUTVA and SITA by construction, such that the treatment effect is simply the difference between the average outcome for units assigned to the treatment condition and the average outcome for units assigned to the control condition. Defining $E[Y(T = 1)]$ as the expected outcome of units assigned to the treatment condition and $E[Y(T = 0)]$ as the expected outcome of units assigned to the control condition, the average treatment effect (ATE) is defined as:

$$\tau_{ATE} = E[Y(T = 1)] - E[Y(T = 0)]. \quad (2)$$

τ_{ATE} represents the difference between the expected potential outcomes across all units in the sampling frame. Put another way, the ATE estimand is the average effect we would expect to observe for *any* (randomly selected) unit.

In an observational study lacking random assignment, an estimated causal effect is likewise an average. However, because random assignment is not present, SITA is unlikely to hold as units' treatment conditions are likely determined by the units' perceptions of the

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expected outcomes under each condition. Put another way, the assignment to treatment is confounded with the potential outcomes and the characteristics of the units. Morgan and Winship (2015) note that a naïve estimate of the treatment effect is simply the difference between the expected outcome for units in the treatment condition, $E[Y(T = 1)|T = 1]$, and the expected outcome for units in the control condition, $E[Y(T = 0)|T = 0]$, or:

$$\tau_{NAIVE} = E[Y(T = 1)|T = 1] - E[Y(T = 0)|T = 0]. \quad (3)$$

Morgan and Winship (2015) provide a detailed discussion of why this is a poor estimate of the ATE. For our purposes, we simply note that the bias in τ_{NAIVE} increases in the extent to which units self-select into treatment conditions based on their perceptions of the potential outcomes.

Researchers performing observational studies attempt to address this self-selection bias by estimating a causal effect as the difference in an outcome of interest between a treatment sample and a control sample matched (or reweighted) based on observable covariates. The resulting estimand constructed based on this matched sample design is an average treatment effect of the treated (ATT), representing the effect of the treatment for *those observations that actually received the treatment*. Stated formally, the ATT is:

$$\tau_{ATT} = E[Y(T = 1)|T = 1] - E[Y(T = 0)|T = 1], \quad (4)$$

where $E[Y(T = 1)|T = 1]$ is the expected potential outcome for treated units that actually receive the treatment, which is readily observable from the data, and $E[Y(T = 0)|T = 1]$ is the expected potential outcome for treated units if they do *not* receive the treatment. This latter expectation involves a potential outcome that is unobserved and must be estimated using the matched control sample of observations.

In observational studies, an estimate of the ATT is not immediately generalizable to the treatment effect for all observations in the sample (in particular, those *not* receiving the

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treatment). This point is subtle, but important. For example, the causal effect of a Big-N audit is often estimated by identifying a matched or weighted control sample of firms that, although not audited by a Big-N auditor, are similar in underlying fundamental characteristics (e.g., firm size) to Big-N audit clients. In this case the ATT reflects the effect on audit quality of a Big-N audit for firms that are typically audited by Big-N auditors (e.g., larger firms). This estimated effect is not readily generalizable as the causal effect of a Big-N audit for firms that are *not* typically audited by Big-N audit firm, given systematic differences in the client firms of these different auditors, which may drive differences in the treatment effect across these samples.

A second estimand available to empirical researchers attempting causal inference is the average treatment effect on the control (ATC). The ATC is the effect of the treatment for those observations that did *not* receive the treatment. Stated formally, the ATC is:

$$\tau_{ATC} = E[Y(T = 1)|T = 0] - E[Y(T = 0)|T = 0], \quad (5)$$

where $E[Y(T = 1)|T = 0]$ is the expected potential outcome for control units that actually receive the treatment, and $E[Y(T = 0)|T = 0]$ is the expected potential outcome for control units if these units *do not* receive the treatment, the latter of which is readily observable from the data. Researchers estimate an ATC by calculating the difference in an outcome variable across a treatment sample that is matched (or weighted) to be similar to the control sample and the sample of control observations. For example, the ATC estimand is the effect on audit quality of a Big-N audit for firms that are *not* typically audited by a Big-N auditor (e.g., smaller firms).

Closely related to estimates of the ATT and ATC are analyses where researchers isolate subsamples of observations where the assumptions required for causal inference are more tenable. (See Imbens 2010 for a discussion of this point). These designs include stratification, regression discontinuity, and subsamples identified based on institutional features or theory.

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Each of these estimands may be accompanied by a 'local' prefix (i.e., LATE, LATT, or LATC), as the estimated treatment effect is specific to a particular subset of the full sample and not necessarily generalizable to the population.

Entropy Balancing to Reweight Sample Observations to Ensure Covariate Balance

Matched-sample research designs provide advantages over traditional regression-based designs, primarily by avoiding an assumption of a linear relation between an outcome of interest and underlying covariates and by limiting the extrapolation of estimates outside the distributional support of the combined treatment and control samples. This latter regression-based assumption that model parameters remain constant (in a linear fashion) beyond the support of the data can lead to biased inferences and measurement error when OLS methods are applied to observations with extreme values (Cochran and Rubin 1973).

However, matched-sample designs are not without drawbacks. In particular, finding an exact match for each treatment sample observation is difficult (if not impossible) in the presence of multiple covariates, especially when covariates are continuous. For this reason, researchers often use a bandwidth approach when matching to allow for differences between treated and matched control observations. However, imprecision in the match induces noise, which can lead to lower power tests while not necessarily reducing bias in an estimated effect.

Entropy balancing (EB) marks an attempt to overcome the drawbacks of existing matching approaches when identifying a weighted control sample that appears similar to the treatment sample of interest (Hainmueller 2012). In particular, EB focuses directly on estimating non-negative weights for each control sample observation such that the specified moments (e.g., mean, variance, and/or skewness) of the covariate distributions of the weighted control sample are nearly identical to those of the treated sample. This approach ensures covariate balance

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between treatment and control samples without the need to specify a propensity score model or identify a (potentially noisy) one-to-one match for each treated observation. Moreover, Hainmueller (2012) explains that EB qualifies as an "equal percent bias reducing" matching method, which guarantees that bias after matching will be lower than before matching.

To illustrate how EB works empirically, we construct a simulated "toy" example to compare observational weights and estimates across OLS, PSM, and EB approaches.[7] We construct a control sample of 100 observations denoted by an indicator for treatment, $t = 0$, where each observation has a value for a single covariate, $X = U(0, 10)$, where $U(a, b)$ represents the continuous uniform distribution ranging from a to b. Similarly, we construct a treatment sample of 50 observations with $t = 1$ and with $X = U(5, 10)$, such that the range of covariate values for the treatment sample is restricted to the upper half of the distribution present for the control sample. This is analogous to a treatment sample composed of only large firms (with size measured by X), while the control sample comprises both small and large firms. The outcome variable for both treatment and control observations with values of $X > 5$ is defined as: $Y = 0.5 + 0.5 * X + t + \epsilon_1$, where ϵ_1 is a random variable from a normal distribution with a mean of zero and a standard deviation of 0.25. For observations with $X < 5$: $Y = 3 + \epsilon_2$, where ϵ_2 is a random variable with the same distribution as ϵ_1. As a result, treatment observations (with $t = 1$) have a Y value that is 1.0 unit greater, on average, than corresponding observations (with the same X value) in the control sample, such that the ATE is 1.0 in expectation.

Using this simulated dataset, we begin by estimating an OLS regression with a dependent variable of Y and two independent variables: the indicator for treatment, t, and the covariate, X. Because OLS assigns an equal-weight (implicitly) to each observation included in the regression,

[7] For details of this simulation, including State code and figures that summarize covariate balance and observational weights, please see the Online Appendix to this article available at: [Insert Appendix URL here.]

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the resulting estimated coefficient on the indicator for treatment is significantly below the 1.0 ATE (*t* coefficient = 0.817). This downward bias in the estimate of the treatment effect is due to the inclusion of control sample observations with *X* values that lie outside the distribution of values in the treatment sample ($X < 5$) and that contain a differing process for generating the outcome variable *Y*, such that the coefficient estimate on *t* is biased significantly toward zero.

We use the same dataset to implement PSM by estimating a Logit propensity score model that regresses the indicator for treatment, *t*, on the *X* covariate before matching observations without replacement using a caliper of 0.03, resulting in a sample of 92 matched observations (46 treatment and 46 control observations). Examining the distribution of matched observations reveals that PSM eliminates observations from the control sample with *X* values that lie below the minimum value (i.e., 5) observed for the treatment sample. The resulting coefficient estimate on *t* from an OLS regression over this matched sample is insignificantly different from the 1.0 ATE (*t* coefficient = 0.963), consistent with PSM eliminating bias stemming from assuming a constant linear functional form for the relation between *t* and *Y* across the full range of *X* values.

Turning to EB shows that weights assigned to control sample observations differ from those assigned by PSM. While PSM assigned a weight of one to 92 matched observations and a weight of zero (excluded) to the remaining unmatched observations, EB assigns a continuous weight ranging from effectively zero (i.e., 1.27e-18) to 1.56. Examining the distribution of these weights shows that EB assigns positive weights above 1.0 in magnitude to control sample observations with *X* values above 5 in order to equalize the distribution of the *X* covariate across the two samples. In contrast, control sample observations with *X* values below 5 receive weights less than 1.0 and that are essentially zero for all *X* values below 4 in magnitude. The resulting weighted OLS regression using weights identified by EB shows an estimated coefficient on the

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treatment indicator that is insignificantly different from the 1.0 ATE (*t* coefficient = 0.969), in line with results from using PSM. As a result, while both PSM and EB effectively eliminate downward bias in the estimated treatment effect in this simulated example, weights assigned by EB allow for the inclusion of control sample observations with a broader range of *X* values and the inclusion of all 50 treatment observations (with a regression weight of 1.0).

III. BIG-N AUDITS: INSTITUTIONAL BACKGROUND AND DATA DESCRIPTION

Big-N Audit Literature

DeFond and Zhang (2014) comprehensively survey the Big-N audit literature. These authors note that while the bulk of research in this area finds that Big-N audits result in higher audit quality, recent studies that apply matching approaches to address covariate imbalance find evidence to the contrary. Lawrence, Minutti-Meza, and Zhang (2011) use PSM to match firms audited by Big-N auditors to those with non-Big-N auditors and find no difference in three audit-quality proxies across the matched samples. These authors state (p. 259), "We caution the reader that this study has not resolved the question, although we hope that it encourages other researchers to explore alternative methodologies that separate client characteristics from audit-quality effects." Spurred by Lawrence et al.'s (2011) results, DeFond, Erkens, and Zhang (2017) vary the PSM analysis choices made by Lawrence et al. (2011) in 3,000 ways and document evidence supporting the presence of a Big-N effect in other audit quality proxies. Taking the approach of estimating a local average treatment effect (LATE), Jiang, Wang, and Wang (2019) document that firms switching to Big-N auditors due to plausibly exogenous shocks stemming from Big-N audit firm acquisitions experienced an increase in audit quality. Given this stream of literature, our study is methodological in nature and our intent is not to contribute to the literature on a Big-N effect, but merely to use this well-documented setting as a useful empirical

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illustration of several pitfalls common to using EB as a tool for addressing covariate imbalance.[8]

Arthur Andersen Demise

Given differences in the underlying client firms served by Big-N and non-Big-N auditors, we examine the audit fee premium in a setting with two features that are common to a wide range of studies in empirical-archival accounting research and are important for outlining potential pitfalls with EB. In particular, we construct pooled (cross-sectional) and difference-in-differences estimates of the audit fee premium around a watershed event in the U.S. audit market by focusing on the events surrounding the demise of Enron and its auditor, Arthur Andersen. On October 16, 2001, Enron announced a $1.01 billion asset write-down. This largely unanticipated announcement was followed relatively quickly by the firm's bankruptcy filing on December 2, 2001. In the aftermath of the large loss in equity market capitalization for Enron, Congress opened an investigation into Enron's failure and the role of Arthur Andersen. These events culminated in Congress signing the Sarbanes-Oxley Act into law on July 30, 2002, quickly followed by the cessation of audits by Arthur Andersen on August 31, 2002.[9]

We use this setting to construct a panel dataset in the periods surrounding the 2002 event year of the shock to the U.S. audit market outlined above. Using this dataset, we examine whether Big-N audit firms command a fee premium for their audits, suggesting a Big-N auditor reputation premium surrounding the demise of Arthur Andersen.[10]

Audit Fee Model: Data and Estimation

[8] We also caution readers that analysis techniques for observational data, including matching, do not obviate the need for addressing underlying assumptions (i.e., SUTVA, SITA) when drawing a causal inference (Rubin 2008).
[9] For a detailed discussion of the timeline of events surrounding the collapse of Enron and Arthur Andersen, see Krishnamurthy, Zhou, and Zhou (2006, p.469).
[10] Studies examining equity market reactions in response to key event dates for Arthur Andersen client firms during this period are mixed regarding a priced auditor reputation effect, with debate centering on whether equity returns reflect auditor reputation for client firms (e.g., Krishnamurthy, Zhou, and Zhou 2006) or merely stem from exposure to oil price shocks given Arthur Andersen's client base (e.g., Nelson, Price, and Rountree 2008) during this period.

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We construct a sample of firm-year observations with data available to estimate the audit fee model specified in DeFond and Zhang's (2014) Table 3, which includes data on financial determinants from Compustat and data on the audit firm, fee, and opinion from Audit Analytics (both databases available via Wharton Research Database Services [WRDS]). To examine whether Big-N auditors command a fee premium from their clients and whether this premium changes around the demise of Arthur Andersen in August 2002, we limit the sample to fiscal years ending between June 2000 and May 2005. We begin in 2000 as this is the year that Audit Analytics begins full coverage of Compustat firms. Because we omit fiscal year 2002 in the bulk of our empirical tests as a transition year that immediately follows the collapse of Enron (filing for bankruptcy in December 2001) and that includes the cessation of audits by Arthur Andersen (in August 2002), we use a pre-period of fiscal years 2000 – 2001 and a post-period of fiscal years 2003 – 2004, resulting in balanced pre- and post-event periods.

Table 1 presents the annual number of observations for firms with and without a Big-N auditor. In Panel A we present the annual number of observations in our full sample, while in Panel B we report the annual number of observations classified as non-accelerated filers in 2004, a status granted by the Securities and Exchange Commission to smaller firms that permits additional time in filing the required 10-K annual report. Table 1 shows that our full sample of firm-year observations is tilted toward large firms with a Big-N auditor, with roughly 5-6 times as many Big-N client firms in the pre-period (2000 – 2001). While this imbalance in sample counts declines in the post-period (2003 – 2004) to roughly 2-3 times the number of non-Big-N (control) sample observations, our full sample continues to display a relatively larger pool of Big-N client firms. In contrast, Panel B shows that the numbers of observations for the non-accelerated filers subsample are more balanced, with more (fewer) observations with a Big-N

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auditor in the pre- (post-) period. This sample composition aligns with evidence that Big-N audit firms tended to drop smaller clients in response to deliberations surrounding the Sarbanes-Oxley Act and the resulting demise of Arthur Andersen (Landsman, Nelson, and Rountree 2009).[11]

Using the samples detailed in Table 1, we begin by estimating the audit fee premium for a Big-N audit using the pooled sample of observations across the 2000 – 2004 sample period, reflecting the typical use of EB to construct a cross-sectional estimate of a treatment effect of interest. We estimate both an ordinary least squares (OLS) regression and a weighted OLS regression, where weights for observations are identified using either PSM or EB:

$$\begin{aligned} ln(Fee_{it}) = {} & \alpha + \beta_1 BigN_{it} + \beta_2 Size_{it} + \beta_3 Leverage_{it} + \beta_4 Loss_{it} + \beta_5 ROA_{it} + \\ & \beta_6 Cur_Ratio_{it} + \beta_7 Quick_Ratio_{it} + \beta_8 ForSales_{it} + \beta_9 ln(BusSeg_{it}) + \\ & \beta_{10} DecFYE_{it} + \beta_{11} Going_Concern_{it} + \gamma_i + \varepsilon_{it}. \end{aligned} \quad (6)$$

The dependent variable is the natural logarithm of the annual audit fee, $ln(Fee_{it})$, obtained from Audit Analytics. Our primary independent variable of interest is an indicator variable for whether a firm obtains a Big-N audit, *BigN*, set equal to one for firms with an auditor with a rank of 1 - 6 in Audit Analytics for fiscal years ending in 2000 or later, which comprises audits by Arthur Andersen, Deloitte, Ernst and Young, KPMG, and PwC (and their predecessor firms), and to zero otherwise. In cases where multiple audit firms performed work during the year (due to a change in auditor, for example) we retain the audit firm that ultimately issued the audit opinion for the fiscal year (and the resulting audit fee) for purposes of our analysis.

In addition to the cross-sectional estimate in Eq. (6), we also estimate a model to allow for expected variation in the audit fee premium surrounding the demise of Arthur Andersen:

$$\begin{aligned} ln(Fee_{it}) = {} & \alpha + \beta_1 BigN_{it} + \beta_2 PostAA_t + \beta_3 BigN_{it} \times PostAA_t + \beta_4 Size_{it} + \beta_5 Leverage_{it} + \\ & \beta_6 Loss_{it} + \beta_7 ROA_{it} + \beta_8 Cur_Ratio_{it} + \beta_9 Quick_Ratio_{it} + \beta_{10} ForSales_{it} + \\ & \beta_{11} ln(BusSeg_{it}) + \beta_{12} DecFYE_{it} + \beta_{13} Going_Concern_{it} + \gamma_i + \varepsilon_{it}. \end{aligned} \quad (7)$$

[11] To address changes in sample composition over this period, our subsequent tests explore requiring client firms to be present in the sample as of 2001 (the last pre-event period). See Section 5 for details.

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PostAA is an indicator set to one for fiscal years after the demise of Arthur Andersen (2003 – 2004) and to zero for pre-demise years (2000 – 2001). Our difference-in-differences estimate of the audit fee premium is captured by the β_3 coefficient on the *BigN*×*PostAA* interaction.[12]

Remaining control variables in Eqs. (6) and (7) reflect fundamental determinants that prior research uses to capture complexity and risk underlying the pricing of a firm's audit. These include the firm's market capitalization (*Size*), debt-to-assets ratio (*Leverage*), an indicator for years with negative income before extraordinary items (*Loss*), return on assets (*ROA*), measures of liquidity via the current ratio (*Cur_Ratio*) and quick ratio (*Quick_Ratio*), the proportion of total sales made to foreign countries (*ForSales*), the natural log of the number of business segments reported (*ln[BusSeg]*), an indicator for a December fiscal year-end (*DecFYE*), and an indicator for firms receiving a going concern opinion indicating financial distress (*Going_Concern*). We winsorize all continuous, unlogged variables at the top and bottom percentiles to limit effects of potential outlying observations and data errors in our analysis. We also include 2-digit SIC industry fixed effects (γ_i) in the estimation.[13] Appendix A provides detailed variable definitions and data sources for all variables used in our tests.

IV. EMPIRICAL ANALYSIS OF THE BIG-N AUDIT FEE PREMIUM

Covariate Imbalance

As part of our estimation of the audit fee models in Eqs. (6) and (7), we begin by

[12] In constructing the difference-in-differences estimate, we implicitly assume that the *PostAA* variable is the result of an event that is plausibly exogenous with respect to individual client firms. If *PostAA* is not exogenous, we face a "double-treatment" problem, where the construct of interest is the interaction between two endogenous treatment variables. In other research designs where there are likewise two treatment variables of interest (e.g., a Big-N auditor and audit committee expertise), one either must assume that one treatment variable is plausibly exogenous or use a research design that addresses imbalance with respect to both treatment variables. Options here include a 2x2 partition of the sample or extensions designed to apply entropy balancing to settings with a continuous treatment variable (e.g., Vegetabile et al. 2021; Tübbicke 2020). We thank an anonymous referee for making this point.

[13] Including fixed effects in an EB model is conceptually similar to including fixed effects in a standard OLS model. In both cases, a fixed effect removes the mean value of the outcome variable for the group of (weighted) observations to which the fixed effect applies (e.g., a particular firm or year).

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assessing whether our control sample of firms with a non-Big-N auditor (*BigN* = 0) serves as a valid counterfactual for the treatment sample of Big-N audit clients (*BigN* = 1) based on the extent of covariate balance in observable determinants across these samples. Table 2 presents summary statistics for standardized differences in means and variance ratios for our full sample (see Panel A of Table 1) to assess differences in the first and second distributional moments.

Analysis of Table 2 shows that firms employing a Big-N auditor tend to be substantially larger (based on *Size*), more complex (based on *ForSales* and *ln[BusSeg]*), better performing (based on *ROA* and *Going_Concern*), and carry lower financial leverage (based on *Leverage*, *Cur_Ratio*, and *Quick_Ratio*) on average relative to the pool of firms with a non-Big-N auditor. This covariate imbalance is displayed graphically in Figure 1. Consistent with significant imbalance, standardized differences in means across Big-N and non-Big-N samples are uniformly outside of the bounds suggested by Rubin (2001) for a balanced covariate in Panel A of Figure 1. Similarly, variance ratios presented in Panel B of Figure 1 show that six of the ten covariates included in the audit fee models in Eqs. (6) and (7) also display significant imbalance in the second distributional moment. This raises a concern that the unconfoundedness portion of the SITA assumption is likely violated for any causal effect estimated over these samples.

To address covariate imbalance across Big-N and non-Big-N samples, we employ EB and PSM to reweight observations to achieve covariate balance. To implement EB, we reweight non-Big-N (control) sample observations to equalize the first (mean) and second (variance) moments of the distributions of the ten covariates in the audit fee models (using the default tolerance of 0.015). Table 2 shows that EB eliminates imbalance in both means and variances, evidenced by standardized differences near zero and variance ratios near one for all covariates.

To implement PSM, we first measure propensity scores (p-scores) as the probability of

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treatment estimated from a logit model with the ten covariates in the audit fee models as independent variables. Table 3 presents results of estimating p-scores. Consistent with the pronounced covariate imbalance in client firm size across Big-N and non-Big-N auditors prior to matching, *Size* is the most significant covariate in the p-score model. Further, the p-score model displays strong explanatory power for the decision to engage a Big-N auditor, with an area under the receiver operating curve (ROC) of 0.882. Using the estimated p-scores, we follow Shipman et al.'s (2017) recommendations to match observations without replacement using a common support and caliper of 0.03, resulting in a sample of 5,558 matched observations (2,779 observations from both the treatment and control samples). Figure 1 shows that for the sample of observations matched using PSM, means and variances are within the acceptable bounds for balanced covariates (Rubin 2001). We conclude that in the audit fee setting that we examine, both EB and PSM are successful in addressing covariate imbalance.

Despite EB's success in eliminating imbalance for our full sample of treatment (Big-N clients) and control (non-Big-N clients) observations, Table 2 suggests underlying issues with the weights assigned by EB in this setting. In particular, the bottom rows of Table 2 show that the maximum observational weight that EB assigns to a control observation is 434.81; in other words, a single observation receives roughly 435 times the weight it would receive in an OLS regression. Relative to our Big-N treatment sample of 12,915 firms, this maximum observational weight means that a single non-Big-N client firm effectively serves as the selected counterfactual for 3.4 percent of the treatment sample. Further, the *Weight Ratio,* computed as the number of control (non-Big-N) observations receiving above equal-weight in the EB program scaled by the total number of control observations, shows that only 18 percent of the control sample is up-weighted to achieve balance. Excessive reliance on a single observation or on a small portion of

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the control sample may result in estimates that are sensitive to the idiosyncrasies of these observations. We revisit both diagnostic measures in our subsequent tests to determine the reliability of EB estimates of the audit fee premium that rely on the full sample of observations.

Regression Results: Baseline Regression Models Using the Full Sample of Observations

In Panel A of Table 4, we present results of estimating Eq. (6) on our pooled sample of observations (with 2002 omitted as a transition year) using OLS in model (1), an EB-weighted regression in model (2), and an OLS regression using observations matched using PSM in model (3). Results in Panel A show that the cross-sectional estimate of the Big-N audit fee premium via the β_1 coefficient on the *BigN* indicator variable is positive and statistically significant at the on percent level across models (1) – (3). However, the estimated magnitude of the Big-N fee premium varies substantially across the three models, with the EB estimate in model (2) suggesting a smaller audit fee premium of 29.7 percent (i.e., $e^{0.260} - 1$) relative to both the OLS estimated premium of 51.9 percent and the PSM estimated premium of 49.8 percent.[14]

In Panel B of Table 4, we present results of our difference-in-differences estimate from Eq. (7) of the audit fee premium surrounding the demise of Arthur Andersen. Results for the OLS estimate in model (1) show an insignificant β_1 coefficient on the *BigN* term, consistent with a limited fee premium prior to 2002, combined with a β_3 coefficient on the *BigN*×*PostAA* term that suggests a 72.6 percent greater fee premium for a Big-N audit in the post-period. Taken at face value, this linear estimate predicts a substantial fee premium for a Big-N audit in response to audit quality concerns following the collapse of Enron and its aftermath.

[14] In tests that are untabulated for brevity (available upon request), we also use a simple program (implemented in Stata) to bootstrap these EB estimates by taking the following steps: (1) randomly sample observations from both treatment and control samples with replacement, (2) run the EB procedure on each random sample, and (3) run a survey-weighted regression storing the coefficient from each iteration. This procedure results in nearly identical coefficient and standard error estimates to those tabulated.

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Results in models (2) and (3) of Panel B suggest this inference is at least partially susceptible to functional form misspecification in the OLS model. In particular, while model (2) shows an insignificant β_1 coefficient on the *BigN* term, consistent with evidence in model (1) of a limited fee premium prior to 2002, the β_3 coefficient on the *BigN*×*PostAA* interaction shows a more limited increase of 26.9 percent in the fee premium for a Big-N audit in the post-period. This difference-in-differences estimate for the audit fee premium is less than 40 percent of the OLS estimate in model (1). Turning to results in model (3) for PSM shows that in contrast to models (1) and (2), the estimated coefficient on the *BigN* main effect is significantly positive, consistent with a Big-N audit fee premium of 30.3 percent in the pre-2002 period. Further, results in model (3) also show a significant difference-in-differences estimate for the increase in the fee premium, with a β_3 coefficient on the *BigN*×*PostAA* term showing an increase of 23.5 percent. Taken together, these initial results show that adjusting for covariate imbalance across Big-N and non-Big-N samples is important when estimating the audit fee premium, in line with stark differences in the fundamentals of audit client firms (e.g., Figure 1). We turn next to assessing whether our EB estimates in Table 4 are subject to common pitfalls when using EB on samples with either (or both) relatively limited distributional overlap or a panel structure.

V. COMMON PITFALLS IN ENTROPY BALANCING: EMPIRICAL ILLUSTRATION

Assessing Weights Applied via Entropy Balancing and Dealing with Large Weights

The maximum weight reported in Table 2 shows that using EB to achieve balance on the first two distributional moments for the full sample of firm-year observations results in large weights assigned to a handful of control sample (non-Big-N) observations. Assigning large weights to control sample observations is intuitively similar to sampling with replacement, in which the same control sample observation is selected multiple times as a match. The end result

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is an estimate of the missing potential outcome and the corresponding treatment effect that relies on a handful of control sample observations that are "selected" frequently. This raises a concern with the reproducibility of the estimate of the treatment effect in alternative samples.

The assignment of large weights via EB is more likely when the control sample of observations is relatively small, when the number of balancing conditions (i.e., covariates and distributional moments to be balanced) is large relative to the number of control observations, or substantial covariate imbalance exists between treatment and control samples (or a combination of these factors). Our samples of Big-N and non-Big-N client firms display substantial covariate imbalance with respect to several covariates included in the audit fee model in Eqs. (6) and (7). To further illustrate this imbalance, Figure 2 plots overlapping histograms of the distribution of client firm *Size* for Big-N (in blue) and non-Big-N (in gray) samples. This figure shows that among the full sample of observations, Big-N clients tend to be significantly larger on average than non-Big-N client firms. More importantly, there are very few non-Big-N firms that are large enough to overlap with the mean of the Big-N client firm size distribution. The result of this limited distributional overlap is that a handful of the largest non-Big-N clients (in the control sample) must be assigned very large weights to balance the mean of *Size* across treatment and control samples. In settings with very limited (or no) distributional overlap in covariates across treatment and control samples, the EB algorithm will fail to converge altogether.

An issue with addressing covariate imbalance in the presence of limited overlap across Big-N and non-Big-N samples is not unique to EB as a form of sample reweighting. In Figure 3, we plot the distribution of p-scores from the model estimated in Table 3 for our full sample of observations. The large mass of estimated p-scores clustered near one in Figure 3 shows that a significant fraction of firms in our sample are estimated to be assigned to the treatment condition

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(receiving a Big-N audit) with near certainty. Estimating causal effects using these observations greatly reduces the tenability of the probabilistic overlap portion of the SITA assumption.

To address this issue with limited distributional overlap, PSM permits a straightforward adjustment by deleting observations with p-scores near zero and one.[15] In contrast, EB provides no such straightforward approach for addressing limited distributional overlap in the underlying treatment and control samples. As a result, we consider three solutions for dealing with extreme weights assigned by EB (or with a convergence issue for the EB algorithm). First, Hainmueller (2012) discusses deleting observations with large observational weights assigned by EB before re-estimating control sample weights, effectively putting a ceiling on the weight assigned to a single observation.[16] In some settings, trimming observations from the control sample will result in EB assigning large weights to a broader set of control observations to achieve covariate balance. However, this redistribution of weights requires a sufficient pool of overlapping control sample observations. In the Big-N setting, the limited number of large non-Big-N client firms means that trimming observations from the control sample is likely to *worsen*, rather than improve, the issue with large weights (which we confirm below). In contrast, researchers may trim observations using an estimated p-score to identify observations unlikely to act as a valid counterfactual for either the treatment or control sample. This latter approach to trimming observations with limited distributional overlap effectively combines PSM and EB approaches.

Second, in settings where the treated sample has a wider distribution of key covariates (or a much larger number of observations) relative to the control sample, the researcher may assign

[15] We estimate a version of our PSM models where we trim observations with propensity scores less than 0.05 and greater than 0.95. We find that this approach to trimming observations with extreme propensity scores results in similar covariate balance across treatment and matched control samples as our baseline PSM samples.
[16] Placing a ceiling on the desired observational weights also aligns with a recommendation by Stuart (2010) that when matching with replacement, weights on individual observations should be monitored by the researcher.

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weights to treated observations rather than to the control sample, by essentially "flipping" the treated and control samples in the EB program. This strategy exploits the fact that matching algorithms are more efficient in cases with a deep pool of observations from which to select a match. In the Big-N setting, we have a much larger pool of observations with a Big-N auditor with a wider resulting distribution of covariates relative to the non-Big-N sample. While flipping the treated and control samples in the EB program has advantages with respect to the efficiency of the observational weights, this approach carries the potential drawback of generating an estimate of the average treatment effect for the control (ATC) rather than an average treatment effect for the treated (ATT). In the audit fee setting, estimating an ATC results in focusing on smaller firms when estimating a treatment effect relative to the typical Big-N client firm.

Third, the researcher may focus attention on some subset of observations that display greater distributional overlap between treatment and control samples, in order to improve the quality of the resulting match and ensure tenability of the SITA assumption. In our setting, very large firms employing a Big-N auditor are unlikely to possess a valid counterfactual in the non-Big-N control sample, as it is unlikely that a non-Big-N firm would have the resources or capacity necessary to conduct the annual audit of any of these large, typically multi-national, firms. To construct an estimate of a *local* treatment effect (LATT), we consider the subsample of firms with non-accelerated filing status in 2004, comprising a sample of smaller firms (with a market capitalization below roughly $250 million) where the choice of a Big-N or non-Big-N audit is plausible.[17] Consistent with stronger distributional overlap for this subsample, Panel B of

[17] An additional benefit of examining non-accelerated filers (from an institutional standpoint) is avoiding confounding effects in audit fees from the anticipated implementation of Section 404 of Sarbanes-Oxley. Under the compliance dates originally established in the act, companies meeting the definition of an "accelerated filer" in Exchange Act Rule 12b-2 would have become subject to the internal control reporting requirements with respect to the first annual report that they filed for a fiscal year ending on or after June 15, 2003.

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Figure 3 shows that estimated p-scores are not clustered near the extremes of zero and one for this subsample. Despite greater distributional overlap, an obvious drawback to this approach is that the resulting LATT estimate may not generalize to remaining sections of the distribution (of client firm size, in our setting).[18]

We implement the three proposed solutions described above for dealing with extreme weights in our audit fee setting. Specifically, we re-run the EB program described in Section 4 (see Table 2) with three alternative adjustments: (1) trimming observations with large weights (identified both via the EB program and via estimated p-scores) before re-running the EB program, (2) assigning weights to the treatment sample of Big-N firms (*BigN* = 1) rather than to the control sample of non-Big-N firms, and (3) using the subsample of non-accelerated filers. Within the latter two approaches, we also explore trimming observations with large weights and re-running the EB program to assess the performance of weight trimming in each setting.

To implement the first proposed solution for extreme weights, we begin by deleting two observations from the control sample of non-Big-N audit clients with weights assigned by EB that are greater than 200 in magnitude (as an admittedly ad hoc threshold for an extreme weight) and re-estimating the EB program. This approach results in a maximum observational weight that *increases* from 434.8 (see Table 2) to 449.9 after trimming. Iterating through this process additional times produces a similar result, where the maximum observational weight continues to climb, suggesting that weight trimming in the control sample is ineffective in addressing extreme weights in the EB program in this setting. Intuitively, this stems from the limited distributional

[18] While the substantial covariate imbalance in client firm size in our setting suggests a relatively straightforward identification of a subsample in which to estimate a LATT, researchers in other settings may elect to examine a quasi-experimental setting (in which observations are effectively randomly assigned to treatment), use a multi-variate approach to select a subsample with greater distributional overlap, or stratify their sample based on a key covariate before estimating a treatment effect within each strata (Frangakis and Rubin 2002).

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overlap between the Big-N and non-Big-N samples, such that there are only a handful of non-Big-N clients that are large enough to achieve covariate balance across these samples when assigning weights to the sample of non-Big-N client firms (see Figure 2). As an alternative, we explore using estimated p-scores to trim observations from both treatment and control samples with p-scores less than 0.05 and greater than 0.95 before re-running the EB program. This process eliminates 5,785 observations (34.3 percent) from the full sample, with 95 percent of eliminated observations coming from the treatment sample and representing large, Big-N audit clients that lack a valid counterfactual in the control sample. In contrast to the approach of trimming observations only from the control sample using weights identified by EB, this trimming procedure is effective in reducing the influence of extreme weights, with a maximum observational weight of 39.9 after implementing EB on this trimmed subsample of observations.

We next move to assessing the remaining two proposed solutions for dealing with extreme observational weights. In Panel A of Table 5, we present summary statistics for covariate balance and observational weights assigned via EB after "flipping" the treated and control samples. In contrast to the baseline EB program, Panel A shows that the resulting sample of weighted *treated* observations now comprises a sample of client firms that are smaller, worse performing, and display higher financial risk. This results from setting EB's target moments using the control sample of non-Big-N clients. Further, this approach results in a larger *Weight Ratio* (0.22 vs. 0.18 in Table 2) and a substantially smaller *Maximum Weight* (53.41 vs. 434.81 in Table 2). These summary statistics lead us to conclude that assigning weights to the larger pool of Big-N observations via EB results in a more efficient set of observational weights.

We also explore whether trimming observations with large weights assigned by EB is an effective strategy for dealing with extreme weights in this alternative ("flipped samples")

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specification. To do so, we eliminate a single Big-N observation with the largest observational weight from the sample and rerun the EB program. While the decision to trim just a single observation is arbitrary, the broad goal of this exercise is to determine whether trimming in this setting allows for any improvement in the efficiency of the EB weights.[19] We find that rerunning the EB program on the trimmed sample results in a new *Maximum Weight* of 30.93 in Panel A, consistent with trimming serving to reduce the maximum observational weight in this setting.

Panel B of Table 5 presents summary statistics for covariate balance and observational weights for the subsample of observations classified as non-accelerated filers in 2004, which we use to construct our estimates of the local average treatment effect for the treated (LATT). Consistent with non-accelerated filers representing a sample of smaller firms, Panel B shows that the treatment sample of Big-N audit clients (*BigN* = 1) has a smaller mean *Size* relative to the full sample in Table 2. Further, the average Big-N client firm in Panel B displays relatively poorer financial performance, evidenced by lower (higher) average *ROA* (*Going_Concern*). More importantly, consistent with greater distributional overlap between Big-N and non-Big-N client firms among non-accelerated filers, we find that using EB on this subsample results in a *Weight Ratio* consistent with more control observations receiving above-equal weight relative to our baseline EB program (0.27 vs. 0.18 in Table 2) and in a substantially smaller *Maximum Weight* assigned to a non-Big-N observation (72.45 vs. 434.81 in Table 2). These summary statistics support the estimation of a LATT to limit issues with extreme weights in the full sample of audit clients. In Panel B, we also examine whether trimming based on observational weight reduces

[19] From a regression diagnostic standpoint, observations with large weights specified by a matching algorithm are likely to act as high leverage points (i.e., outlying observations). For example, Belsley, Kuh, and Welsch (1980, p. 28) suggest that observations with a regression leverage greater than *2k/n*, where *k* is the number of regressors (including the intercept) and *n* is the number of observations, are potential outliers. For our analysis that focuses on covariate balance, we rely on the observational weights rather than regression leverages to avoid requiring the use of the outcome of interest (e.g., audit fees) in identifying the matched sample. Despite this, we find that trimming a handful of observations based on a regression leverage cutoff yields similar results (results untabulated).

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the *Maximum Weight* in the non-accelerated filers subsample. As in Panel A, we trim one observation with the largest observational weight from the sample and rerun the EB program. This results in a slight reduction of the *Maximum Weight* to 65.26.

Given evidence in Table 5 that both strategies designed to reduce extreme observational weights assigned by EB (assigning weights to Big-N rather than non-Big-N firms; excluding accelerated filers) effectively reduce extreme observational weights, we next present results of re-estimating Eq. (7) using the full sample of observations (Table 6) and the subsample of non-accelerated filers (Table 7) when using the adjusted EB programs detailed in Table 5. In both tables, we also include models estimated after trimming observations with extreme p-scores (less than 0.05 and greater than 0.95). Starting first with Table 6 that relies on the full sample of observations, in contrast to our OLS and EB estimates in Table 4, results in Table 6 now show a significant Big-N audit fee premium *prior* to the demise of Arthur Andersen. Results in models (1) and (2) show a β_1 estimate of the pre-period premium of 35 – 36 percent when EB is used to assign weights to the treatment sample of Big-N audit clients. Turning to the PSM estimate in model (3) shows a similar β_1 estimate of the *BigN* pre-period premium of 32.2 percent. Results in model (4) that estimates an EB-weighted regression on the subsample of observations remaining after trimming observations with extreme p-scores (TRIM – EB) shows a lower, but significant positive, β_1 estimate of the *BigN* pre-period premium of 19.2 percent. Further, results in Table 6 show that while difference-in-differences estimates of the Big-N fee premium are statistically significant, these estimates are lower than those for our baseline models in Panel B of Table 4. In particular, models (1) – (4) in Table 6 show an increase in the Big-N fee premium of only 18 – 21 percent based on coefficients on the *BigN*×*PostAA* terms, while coefficient estimates in Table 4 indicate an increase of 23 – 27 percent in the fee premium for EB and PSM models.

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In Table 7, we present results of estimating modified versions of Eq. (7) for the subsample of audit clients classified as non-accelerated filers. Results in model (1) using an unweighted OLS regression show both a significant pre-period Big-N fee premium (β_1 = 0.229, *p*-value < 0.01) and a significant difference-in-differences estimate (β_3 = 0.319, *p*-value < 0.01). Remaining models in Table 7 show that matching approaches uniformly result in a larger estimate of the pre-period Big-N fee premium (with β_1 ranging between 0.284 and 0.375) and a correspondingly smaller difference-in-differences estimate (with β_3 ranging between 0.133 and 0.213). Consistent with results in Table 6, this evidence shows that matching using either EB or PSM approaches results in smaller difference-in-differences estimates for the Big-N fee premium (14 – 24 percent) relative to OLS estimates for the non-accelerated filers subsample (36 percent). Last, we find that the difference-in-differences estimate in the non-accelerated filers setting is somewhat sensitive to the exclusion of one observation with an extreme weight in our EB models, evidenced by a drop in the *BigN*×*PostAA* coefficient when moving from model (2) (coeff. = 0.152, *p*-value < 0.05) to model (3) (coeff. = 0.133, *p*-value < 0.10).

Taken together, results in Tables 6 and 7 show that adjusting for extreme weights in the EB program is important for the resulting estimates of an audit fee premium, consistent with a relatively smaller (and more stable) fee premium for smaller client firms after adjusting for limited distributional overlap across the full sample of Big-N and non-Big-N audit clients. Suggested solutions of estimating an ATC by assigning weights to the thicker pool of treatment (*BigN* = 1) observations or focusing on a LATT estimand (via the subsample of non-accelerated filers) appear effective in addressing issues posed by large observational weights. Further, while we find that trimming observations based on excess weights identified via the EB program is of limited use in our Big-N setting, rerunning EB after trimming observations based on estimated

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propensity scores significantly reduces the influence of large observational weights for both the full sample and the subsample of non-accelerated filers in our empirical setting.

Entropy Balancing in a Panel Setting

The second implementation issue we consider with EB is the assignment of weights to observations in settings where the treatment impacts observations in more than one period. In these settings, assigning weights to the pooled sample of observations via EB will result in assigning weights to control sample observations from both pre- and post-treatment periods in order to achieve balance with the treatment sample (across combined pre- and post-treatment periods). To the extent that there are shifts in the underlying sample of observations across periods or significant changes to the observations themselves, using EB to assign weights across the pooled sample of observations may result in biased estimates of a treatment effect.

For panel data research settings, we note three primary options researchers may consider when balancing covariates: balancing across the entire (pooled) panel, balancing observations in a specific pre-treatment-period, or balancing separately in each period of the panel. These approaches each entail costs and benefits.

Taking first the approach of balancing across the full panel of observations, this approach assumes that each control sample observation (regardless of period) is potentially an appropriate counterfactual for any treatment observation in *either* the pre- or post-treatment period. For example, EB may assign a positive weight to a control observation in the pre-treatment period that appears similar to post-treatment observations in the treatment sample. In this case, while covariate balance may be present for the full panel of treatment and control sample observations, balance is unlikely to be present in the pre-treatment period in isolation. This raises the potential for biased estimates of treatment effects when the treatment is correlated with underlying

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characteristics of the observations. Because we use this pooled approach for our baseline EB estimates (see Table 2), we assess whether alternative approaches to dealing with panel data described next significantly influence our estimates of the Big-N audit fee premium.

The second approach to balancing firms uses the pre-treatment period only and applies observational weights to subsequent periods. This approach entails assuming that control sample weights remain constant in the post-treatment period. This assumption may be problematic if attrition by firms in the control sample and treatment sample differs (thereby generating unbalanced weights for treatment and control samples in the post-treatment period). Further, the assumption of constant weights across periods poses an issue in settings with substantial changes in underlying fundamentals when moving from pre- to post-treatment periods.[20] For example, a study examining pre- and post-crisis periods that balances on pre-crisis covariates is likely to experience substantial imbalance on post-crisis covariates due to marked changes in performance and investment for firms around the crisis, raising the potential for a correlated omitted variable. On the other hand, a benefit of requiring observations to be present in a pre-treatment period is eliminating the effects of shifts in sample composition as a result of the treatment itself. In our setting, requiring observations to be present in the last pre-event year avoids shifts in the client base as a confounding factor when estimating our audit fee models, given a tendency for Big-N auditors to drop smaller clients in response to events surrounding the demise of Arthur Andersen.

A third option is to separately balance observations in each period of the panel, including in post-treatment periods. The advantage to this approach is in ensuring covariate balance while

[20] An issue of diverging characteristics over time between matched treatment and control sample observations is not unique to EB. In other approaches (e.g., PSM or attributes-based matching), researchers must also decide what period to use as a reference to match (e.g., beginning of the panel or the year immediately before the treatment), with the tightness of the match potentially deteriorating over time.

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allowing for sample attrition and time-series variation in covariates.[21] The disadvantage lies in balancing observations on post-treatment variation. To the extent post-treatment variation in covariates is caused by (or related to) the treatment of interest, balancing in the post-treatment period will essentially "balance away" some of the treatment effect of interest.

To empirically illustrate the alternative approaches to dealing with a panel data structure, we rerun our baseline EB program described in Section 4 to balance observations: (1) in the 2001 fiscal year only (i.e., the last pre-treatment year), or (2) separately in each fiscal year of the panel. We present summary covariate balance and observational weighting statistics for these alternative EB programs in Table 8 for our full sample, where Panel A (B) presents results of assigning weights to the control sample of non-Big-N observations in 2001 only (each fiscal year separately).[22] Panel A shows that assigning weights to achieve covariate balance in a single pre-treatment period results in some remaining imbalance on the first moment (via standardized differences in means) and the second moment (via variance ratios) after EB. This is due to the assumption of constant observational weights across all years of the panel, despite shifts in the underlying observations. In contrast, summary statistics in Panel B show that assigning weights separately in each year of the panel results in the elimination of covariate imbalance entirely.

To assess whether assigning weights to individual years in the panel matters for resulting estimates, Figure 4 presents means for logged audit fees in each year of the panel for the treatment sample of Big-N audit clients and for EB-weighted control samples where weights are alternatively assigned to the pooled sample of observations, in 2001 only, and in each year of the panel, respectively. Analysis of Figure 4 shows that the procedure for assigning weights in the

[21] Further, estimating separate EB weights within each time period (e.g., annually) is arguably more theoretically aligned with treatment assignments that can happen in each sample period, such as an auditor appointment.
[22] Requiring observations to be present in 2001 results in 11,505 (2,724) firm-year observations in the full sample and 2,428 (1,872) firm-year observations in the non-accelerated filers sample for Big-N (non-Big-N) audit clients.

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panel dataset matters for the observed trend in audit fees across periods for the weighted control sample. In particular, while the mean audit fee is relatively constant across the 2000 – 2003 period for the non-Big-N control sample using pooled weights (dotted line), mean audit fees display an increasing trend over the same period when weights are assigned in individual years of the panel (dashed lines). More importantly, the trend in audit fees for the weighted control sample aligns more closely with the pre-treatment trend for the treatment sample of Big-N audit clients when weights are assigned in individual years. We conclude from Figure 4 that assigning weights to individual years via EB is likely to result in a less biased estimate of the Big-N fee premium, given the more visible parallel trends in mean audit fees when assigning weights in either the pre-treatment period only or separately in each year of the panel.

In Table 9, we present results of estimating Eq. (7) using weights identified by EB and, as a benchmark, using PSM to identify matched observations based on individual years in the panel. We present results of estimating EB and PSM regression models for our full sample of firm-year observations (with results for the subsample of non-accelerated filers un-tabulated, available upon request). Consistent with univariate evidence in Figure 4, results for EB models in both models (1) and (3) support a smaller difference-in-differences estimate relative to the baseline EB estimate in Table 4 (which uses the pooled sample to assign weights). In particular, results in Table 9 show a difference-in-differences estimate for the Big-N audit fee premium of 16 – 22 percent for the full sample, compared to an estimate of 27 percent in Panel B of Table 4.[23]

Turning to results for trimmed PSM specifications shows that when firms are matched in

[23] Results for EB (PSM) models estimated on the subsample of non-accelerated filers also show modest difference-in-differences estimates, with results using control sample weights identified in the 2001 pre-event period showing an insignificant differences-in-differences estimate of only 2.7 (2.3) percent based on the *BigN×PostAA* coefficient. This suggests that accounting for both shifts in client composition (by requiring each client firm to be present in the 2001 pre-event period) and covariate overlap (by focusing on the subsample of non-accelerated filers) results in an insignificant shift in the Big-N audit fee premium for smaller audit clients around the demise of Arthur Andersen.

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the 2001 pre-event period in model (2), difference-in-differences estimates of the Big-N fee premium are similarly limited, with an estimated increase in the fee premium of 7.6 percent. Turning to PSM estimates constructed in each year of the sample in model (4) shows results that are broadly similar to those in previous tables, consistent with a pooled sample PSM approach performing similar to a yearly match. More broadly, results in Figure 4 and Table 9 suggest that assigning weights to individual years of the panel dataset is important in our audit fee setting for the validity of the parallel trends assumption and for the resulting difference-in-differences estimate of the Big-N audit fee premium, particularly when holding client composition constant.

VI. SUMMARY AND IMPLEMENTATION RECOMMENDATIONS

We empirically illustrate two primary pitfalls when using entropy balancing (EB) in empirical-archival research settings: assigning extreme weights to control sample observations and unevenly assigning weights to pre- and post-treatment periods to achieve covariate balance when using panel data. These pitfalls are pronounced in the audit fee setting that we examine, due to limited distributional overlap in covariates, such as firm size, across treatment and control samples and due to substantial changes in auditor client bases over our sample period. Both of these features appear common in accounting research settings, given the natural focus on estimating treatment effects around significant shifts in the economic and reporting environment. As a result, it appears that while EB offers a number of advantages to researchers when ensuring covariate balance across a treatment and control sample of interest, EB is not without drawbacks. We summarize these primary advantages and disadvantages of EB in Panel A of Table 10.

Our empirical illustration also shows that three proposed solutions for dealing with large observational weights assigned by EB are effective in limiting the influence of these potential outlying observations. In particular, we find that using EB to assign weights to treatment (rather

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than control) observations results in substantially lower observational weights in our setting, with resulting large changes in our estimates of the Big-N audit fee premium. Alternatively, we find that removing observations that are unlikely to possess a valid counterfactual in either the treatment or control sample from the analysis, either by trimming observations with extreme propensity scores or by estimating a local average treatment effect for a sample of treatment and control observations with more substantial distributional overlap, similarly reduces the influence of large observational weights. We also document two proposed solutions for assigning weights to panel data. Specifically, we find that using EB to assign weights to a single pre-treatment year or to assign weights separately within each period in the panel both result in a weighted control sample with more evident parallel trends with the treatment sample in pre-treatment periods.

We summarize these implementation recommendations in Panel B of Table 10. Broadly, we encourage accounting researchers to assess whether extreme observational weights or systematic differences across periods in a panel dataset substantially influence EB estimates (or estimates from alternative matching algorithms, such as propensity score matching) in other settings. On this point, we expect that presenting and discussing maximum observational weights and assessing the distribution of weights within individual sample periods will aid readers and reviewers in evaluating the quality of matched-sample estimates in a range of settings. To this end, we also encourage further empirical work that builds on the tests in this study.

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Appendix A
Variable definitions and data sources

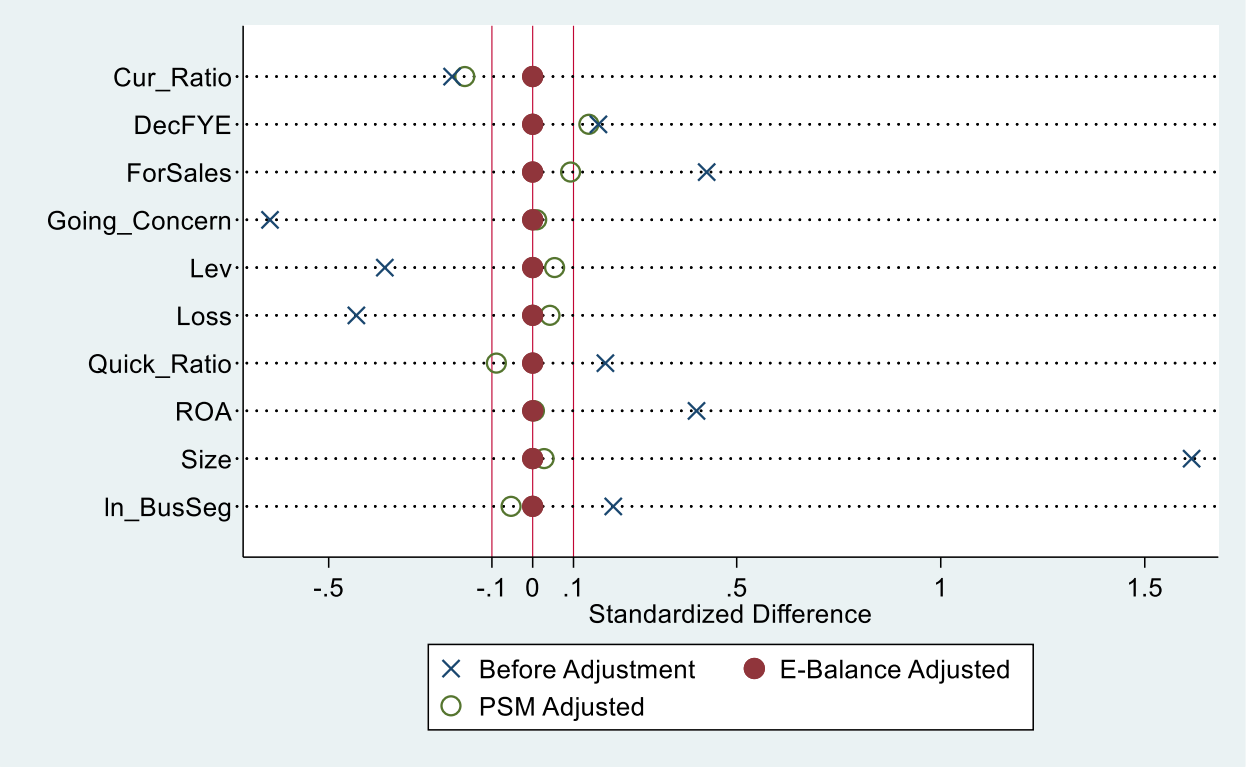
Variable	Description	Data Source
BigN	An indicator variable set equal to one for firms with an auditor with a rank of 1 - 6 in Audit Analytics for fiscal years ending in 2000 or later, covering audits by Arthur Andersen, Deloitte, Ernst and Young, KPMG, and PwC (and their predecessor firms), and to zero otherwise.	Audit Analytics
Cur_Ratio	Current ratio measured as total current assets scaled by year-end book value of total assets.	Compustat
DecFYE	An indicator variable set equal to one for observations with a December fiscal year-end and to zero otherwise.	Compustat
ForSales	The proportion of net sales accounted for by foreign sales, measured using the merged segment database available via Wharton Research Database Services (WRDS) where the geographic segment type = 3 (denoting foreign segments).	Compustat
Going_Concern	An indicator variable set equal to one for observations receiving a going concern opinion in their current annual audit and to zero otherwise.	Audit Analytics
Leverage	The sum of short-term and long-term debt scaled by the year-end book value of total assets.	Compustat
ln(BusSeg)	The natural logarithm of the number of business segments available from the segment type database in Compustat.	Compustat
ln(Fee)	The natural logarithm of annual audit fee.	Audit Analytics
Loss	An indicator variable set equal to one for observations with negative income before extraordinary items and to zero otherwise.	Compustat
PostAA	An indicator variable set equal to one for fiscal years ending in June 2003 or later to reflect audit periods occurring after the August 31, 2002 date on which Arthur Andersen ceased to provide audit services, and to zero otherwise.	
Quick_Ratio	Quick ratio measured as (total current assets - inventory) / total current liabilities.	Compustat
ROA	Return on assets measured as income before extraordinary items scaled by the year-end book value of total assets.	Compustat
Size	The natural logarithm of total market capitalization measured by the book value of total liabilities plus the market value of equity.	Compustat

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Figure 1
Covariate imbalance in the first and second distributional moments across the sample of firm-year observations with a Big-N auditor (treatment) and without a Big-N auditor (control)

This figure presents standardized differences in means and variance ratios for each covariate included in the audit fee model detailed in Eq. (6) for the unadjusted full sample of observations (*Before Adjustment*), after using entropy balancing to achieve balance on the first (mean) and second (variance) moments of each covariate distribution (*E-Balance Adjusted*), and after conducting propensity score matching following the program described in Section 4 where all observations with a valid match, without replacement, based on estimated propensity scores are retained (*PSM Adjusted*). The sample used for the figure aligns with the full sample of observations described in Tables 1 and 2, comprising all observations with available data to estimate Eq. (1) over the 2000 – 2004 sample window (omitting 2002 as the transition year during which Arthur Andersen ceased conducting audits). We use entropy balancing to assign weights to the control sample of firm-year observations without a Big-N auditor (*BigN* = 0) to equalize the first (mean) and second (variance) moments of each covariate distribution with respect to the treatment sample of firms with a Big-N auditor (*BigN* = 1) within the default tolerance of 0.015. The propensity score model and resulting estimates appear in Table 3. Standardized differences are calculated as the difference in means between the treatment and control samples divided by the square root of the average variance in the treatment and control samples for each covariate. The standardized difference will approach zero when the first moment of the distribution for a particular covariate is balanced across treatment and (weighted) control samples. Variance ratios for each covariate are calculated as the ratio of the covariate's variance for the treatment sample scaled by variance for the control sample. Standardized differences and variance ratios outside of the vertical bands in each panel denote differences that indicate significantly imbalanced covariates following the guidance of Rubin (2001) and Austin (2011). See Appendix A for detailed variable definitions and data sources.

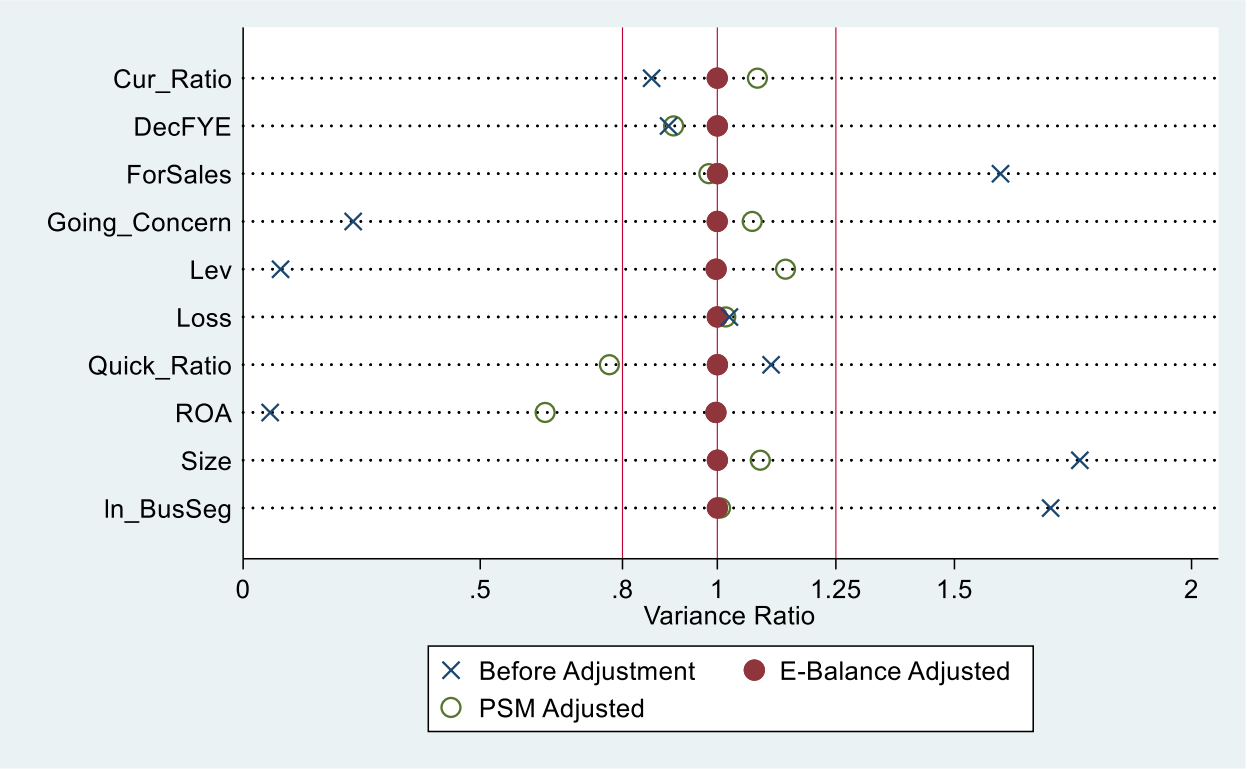
Panel A: Standardized differences in means before and after matching



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Figure 1, continued

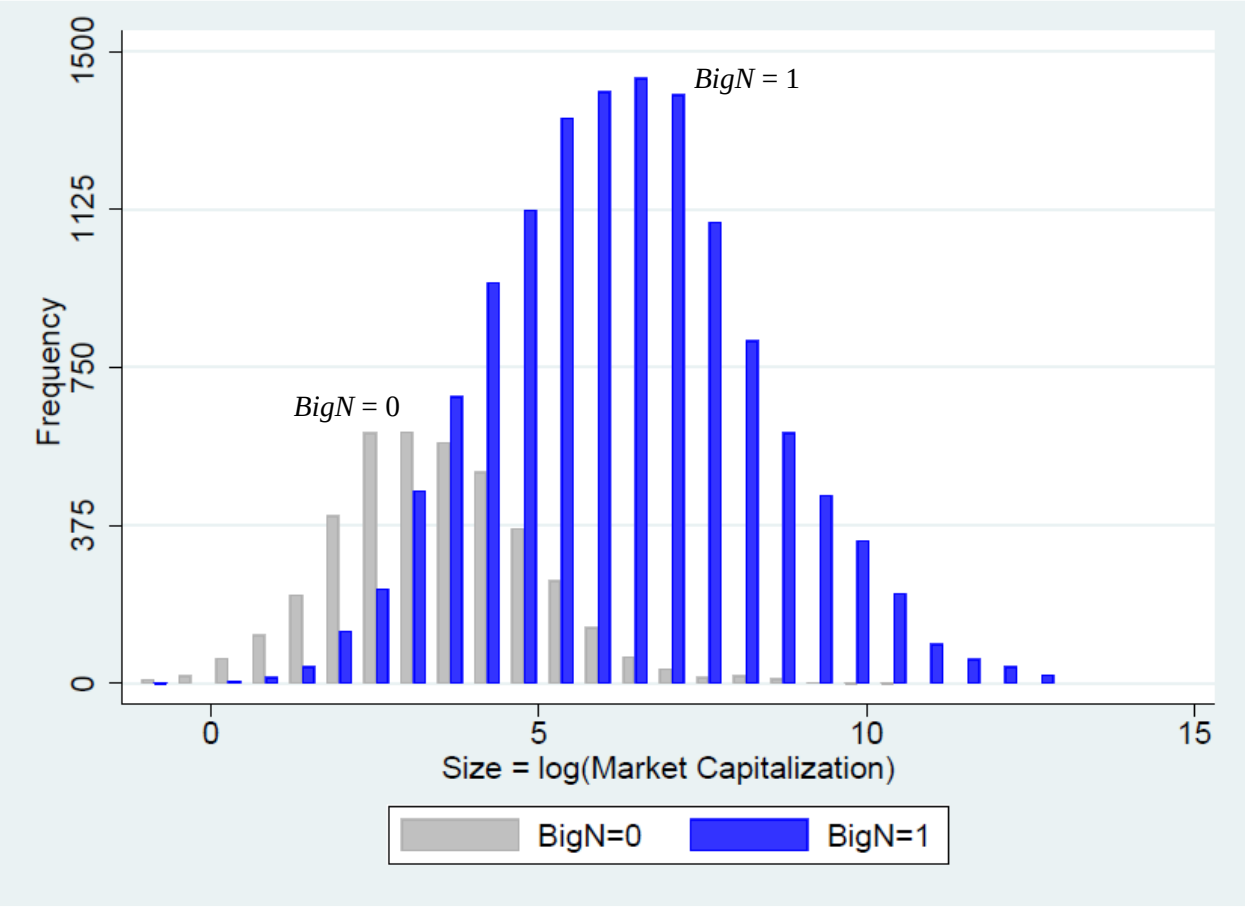
Panel B: Variance ratios before and after matching



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Figure 2
Overlapping histograms of firm size (market capitalization) for samples of firms with a Big-N auditor (treatment) and those without a Big-N auditor (control)

This figure presents overlapping histograms of the frequency distribution of firm size measured by the natural logarithm of total market capitalization, measured by the market value of common equity plus the book value of total liabilities (*Size*). Separate histograms are presented for firm-year observations over the 2000 – 2004 sample window (omitting 2002 as the transition year in which Arthur Andersen ceased providing audits) for firms with a Big-N auditor (*BigN* = 1) and those without a Big-N auditor (*BigN* = 0). Refer to Appendix A for detailed variable definitions and data sources.

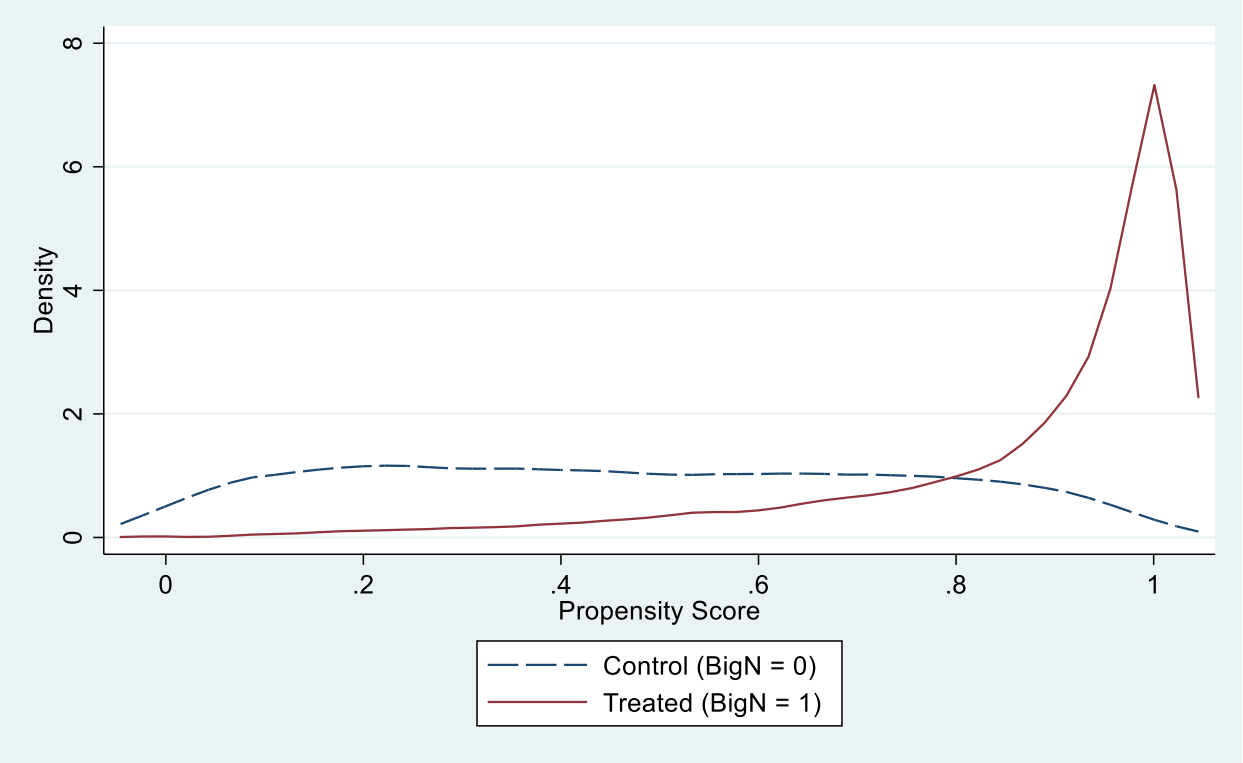


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Figure 3
Propensity score density for Big-N client firms (treatment) and non-Big-N client firms (control)

This figure plots the density of propensity scores estimated using a Logit model regression with covariates drawn from the audit fee model in Eq. (6), with estimated coefficients presented in Table 3. Panel A (Panel B) presents propensity scores for the full sample of firm-year observations (subsample of firms with non-accelerated filing status) over the 2000 – 2004 sample window (omitting 2002 as the transition year in which Arthur Andersen ceased providing audits). Separate lines are presented for firms with a Big-N auditor (solid line for Treated, *BigN* = 1) and those without a Big-N auditor (dashed line for Control, *BigN* = 0). Propensity scores are bounded at 0 (i.e., an observation that is certain to engage a non-Big-N audit firm) and 1 (i.e., an observation that is certain to engage a Big-N audit firm). Refer to Section 4 for details of the propensity score matching (PSM) program and to Appendix A for detailed variable definitions and data sources.

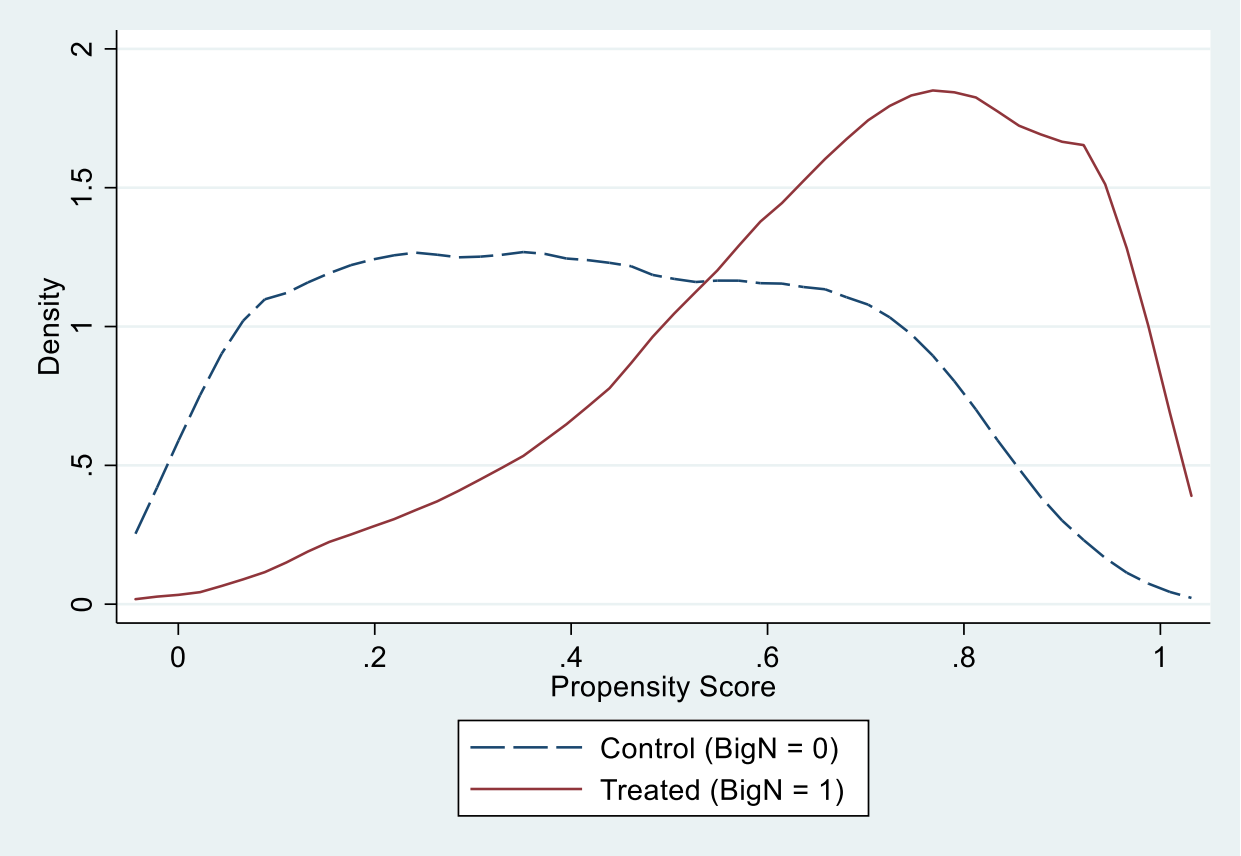
Panel A: Full sample of observations



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Figure 3, continued

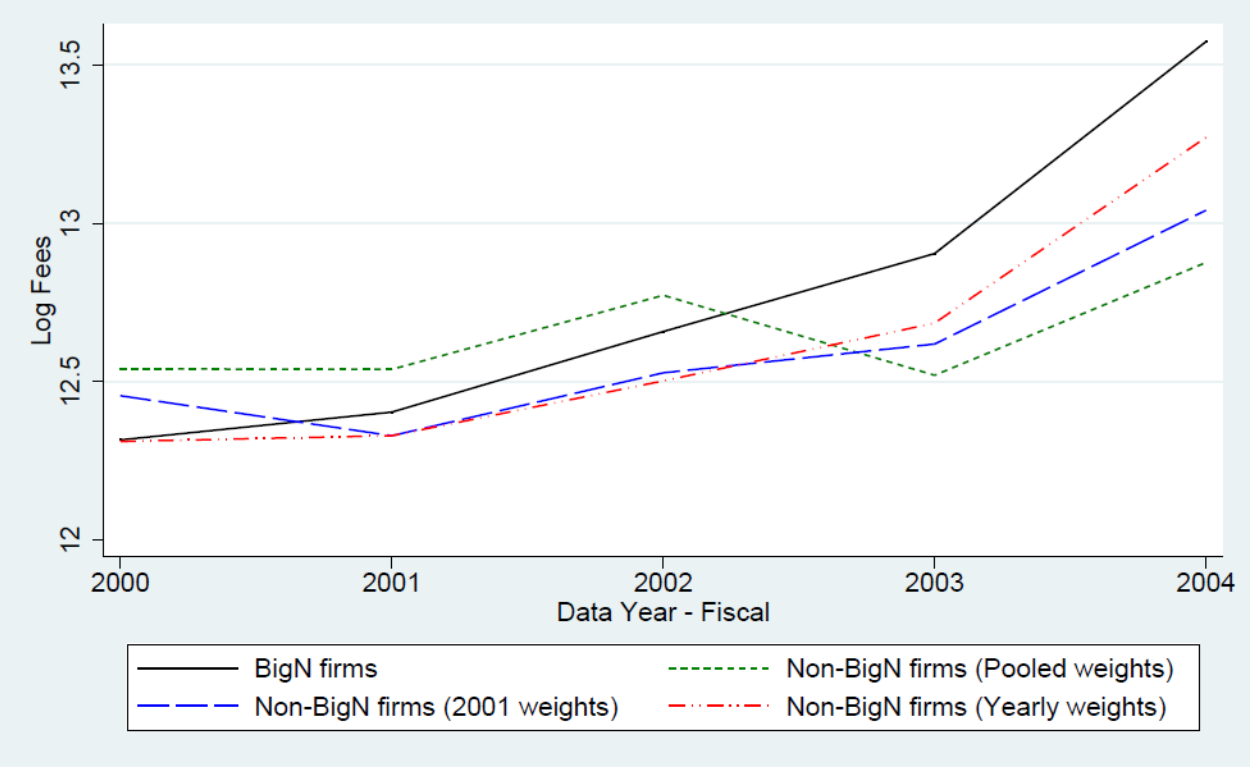
Panel B: Non-accelerated filers subsample



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Figure 4
Mean audit fees by year for firms with a Big-N auditor (treatment) and for entropy-balance-weighted firms without a Big-N auditor (control)

This figure presents means of the natural logarithm of audit fees, *Log Fees*, computed separately by year for four alternative samples of firms: (1) unweighted treatment firms with a Big-N auditor (*BigN firms*), (2) weighted control firms without a Big-N auditor, where weights are assigned by entropy balancing over the full panel dataset in order to balance means and variances with the pooled sample of treatment firms [*Non-BigN firms (Pooled weights)*], (3) weighted control firms without a Big-N auditor, where weights are assigned by entropy balancing to balance means and variances with the sample of treatment firms in the 2001 fiscal year only (as the last pre-event year prior to the demise of Arthur Andersen) [*Non-BigN firms (2001 weights)*], and (4) weighted control firms without a Big-N auditor, where weights are assigned by entropy balancing to balance means and variances with the sample of treatment firms separately in each fiscal year over the 2000 – 2004 sample window [*Non-BigN firms (Yearly weights)*].



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Table 1

Annual frequencies of firm-year observations with and without a Big-N audit firm in the full sample (Panel A) and in the subsample of firms with non-accelerated filing status (Panel B)

This table presents the annual frequency of firm-year observations with data available for the audit fee model in Eq. (1) over the 2000 – 2004 sample window. In Panel A, we present the frequency of observations employing a Big-N auditor (treatment observations, *BigN* = 1) and not employing a Big-N auditor (control observations, *BigN* = 0) in our full sample. In Panel B, we present the frequency of observations after excluding firms with accelerated filing status granted by the U.S. Securities and Exchange Commission (SEC) in 2004 for their 10-K annual report. Please refer to Appendix A for detailed variable definitions and data sources.

	Panel A: Full Sample		**Panel B: Non-accelerated filers**	
Year	**Frequency (BigN = 1)**	**Frequency (BigN = 0)**	**Frequency (BigN = 1)**	**Frequency (BigN = 0)**
2000	3,047	486	633	251
2001	3,512	706	800	418
2002	3,479	1,075	847	782
2003	3,296	1,280	783	1,003
2004	3,060	1,497	776	1,233
Total	**16,394**	**5,044**	**3,839**	**3,687**

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Table 2

Covariate balance statistics before and entropy balancing samples of observations with a Big-N auditor (treatment) and observations without a Big-N auditor (control)

This table presents summary statistics and covariate distributions before and after using entropy balancing to assign weights to firm-year observations without a Big-N auditor (*BigN* = 0) over the 2000 – 2004 sample period (omitting 2002 as the transition period). Standardized differences (*Std Diff*) are presented to assess covariate balance on the first moment (means) between treatment and control samples, computed as the difference in means between the treatment and control samples divided by the square root of the average variance in the treatment and control samples for each covariate. The standardized difference will approach zero when the first moment of the distribution for a particular covariate is balanced across weighted and unweighted samples. Variance ratios are presented to assess covariate balance on the second moment (variance), computed as the ratio of the variance of each covariate in the treatment sample (firms with a Big-N auditor) scaled by variance for the weighted control sample (firms without a Big-N auditor). The *Match Ratio* and *Maximum Weight* are calculated using weights identified for the control sample of observations using entropy balancing to achieve balance on the first and second distributional moments (see McMullin and Schonberger 2020). Refer to Appendix A for detailed variable definitions and data sources.

Prior to balancing:	**Firms with a Big-N auditor (12,915 Observations)**			**Firms without a Big-N auditor (3,969 Observations)**			**Balance Stats**	
Covariate	**Mean**	**Variance**	**Skewness**	**Mean**	**Variance**	**Skewness**	**Std Diff**	**Variance Ratio**
Size	6.267	3.960	0.209	3.423	2.245	0.369	1.615 [a]	1.764 [a]
Leverage	0.229	0.091	6.711	0.515	1.155	4.079	-0.363 [a]	0.079 [a]
Loss	0.409	0.242	0.372	0.620	0.236	-0.494	-0.432 [a]	1.025
ROA	-0.118	0.410	-14.170	-0.901	7.205	-4.616	0.401 [a]	0.057 [a]
Cur_Ratio	0.510	0.063	-0.029	0.562	0.073	-0.221	-0.198 [a]	0.861
Quick_Ratio	2.554	11.530	4.117	1.965	10.360	4.758	0.178 [a]	1.113
ForSales	0.529	0.544	1.459	0.246	0.341	2.866	0.426 [a]	1.597 [a]
ln(BusSeg)	0.981	0.244	0.282	0.894	0.143	1.023	0.197 [a]	1.703 [a]
DecFYE	0.697	0.211	-0.858	0.621	0.235	-0.499	0.161 [a]	0.897
Going_Concern	0.048	0.046	4.205	0.274	0.199	1.014	-0.644 [a]	0.232 [a]
After balancing:								
Covariate	**Mean**	**Variance**	**Skewness**	**Mean**	**Variance**	**Skewness**	**Std Diff**	**Variance Ratio**
Size	6.267	3.960	0.209	6.267	3.960	0.007	0.000	1.000
Leverage	0.229	0.091	6.711	0.229	0.091	3.813	0.000	0.997
Loss	0.409	0.242	0.372	0.409	0.242	0.372	0.000	1.000
ROA	-0.118	0.410	-14.170	-0.118	0.411	-8.901	0.000	0.997
Cur_Ratio	0.510	0.063	-0.029	0.510	0.063	-0.098	0.000	1.000
Quick_Ratio	2.554	11.530	4.117	2.554	11.530	4.448	0.000	1.000
ForSales	0.529	0.544	1.459	0.529	0.544	1.496	0.000	1.000
ln(BusSeg)	0.981	0.244	0.282	0.981	0.244	0.260	0.000	1.000
DecFYE	0.697	0.211	-0.858	0.697	0.211	-0.858	0.000	1.000
Going_Concern	0.048	0.046	4.205	0.048	0.046	4.204	0.000	1.000
Match Ratio	**0.18**	Ratio: Firms with above equal weights in e-balance / All control firms						
Maximum Weight	**434.81**	Maximum weight assigned to a single control sample observation by e-balance						

[a] Indicates covariates with standardized differences (variance ratios) outside of the +/- 0.1 (4/5 and 5/4) bounds suggested by Rubin (2001) as indicating a balanced covariate.

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Table 3
Propensity score model

This table presents estimates of the propensity score for the decision to engage a Big-N auditor estimated using a Logit model for the full sample of firm-year observations over the 2000 – 2004 sample window (with 2002 omitted as a transition year). The dependent variable in the propensity score model is an indicator variable set equal to one for firm-year observations with a Big-N auditor (*BigN*=1) and to zero otherwise. Refer to Appendix A for detailed variable definitions and data sources. t-statistics computed using standard errors clustered by firm appear in parentheses below each coefficient. Area under the receiver operating curve (ROC) is presented as a measure of Logit model fit. *, **, and *** indicate significance at the 10%, 5%, and 1% two-tailed levels, respectively.

Specification	Logit
Dependent Variable	*BigN*
Size	0.914***
	(33.97)
Leverage	-0.544***
	(-5.64)
Loss	0.298***
	(4.63)
ROA	0.024
	(0.88)
Cur_Ratio	0.557***
	(3.80)
Quick_Ratio	0.016
	(1.43)
ForSales	0.128**
	(2.20)
ln(BusSeg)	-0.033
	(-0.39)
DecFYE	0.141*
	(1.94)
Going_Concern	-0.382***
	(-4.30)
Constant	-3.544***
	(-18.71)
Observations	16,884
Area under ROC	0.882

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Table 4

Ordinary least squares (OLS), entropy-balance-weighted (EB), and propensity-score-matched (PSM) regression estimates of the Big-N audit fee premium over the 2000 – 2004 period

This table presents regression estimates of Eqs. (6) and (7) using the full sample of firm-year observations over the 2000 – 2004 sample window (with 2002 omitted as a transition year), where the dependent variable is the natural logarithm of annual audit fees, $ln(Fee_t)$. In Panel A, the independent variable of interest is an indicator variable set equal to one for firm-year observations with a Big-N auditor (*BigN*=1) and to zero otherwise, serving as an estimate of the Big-N audit fee premium for the pooled sample of observation. In Panel B, models include an additional indicator variable set equal to one for fiscal years 2003 – 2004 following the August 2002 demise of Arthur Andersen (*PostAA* = 1) and to zero otherwise, and the interaction between *BigN* × *PostAA* as our difference-in-differences estimate of the Big-N audit fee premium. In both panels, model (1) presents an ordinary least squares (OLS) regression estimate of the audit fee model. Model (2) presents an entropy-balance- (EB-) weighted regression estimate of the audit fee model. Refer to Section 4 for details of the EB program and to Table 2 for covariate balance and observational weighting statistics for the EB model. Model (3) presents propensity-score-matched (PSM) regression estimates using all observations with a valid match, without replacement, after imposing a common support and using a caliper of 0.03. Refer to Section 4 for details of the PSM program and to Table 3 for Logit model estimates of the propensity score. Refer to Appendix A for detailed variable definitions and data sources. All regression models include 2-digit SIC industry-fixed effects. t-statistics computed using standard errors clustered by firm appear in parentheses below each coefficient. *, **, and *** indicate significance at the 10%, 5%, and 1% two-tailed levels, respectively.

Panel A: Pooled estimation of Big-N fee premium over the 2000 – 2004 period

Specification	OLS	EB	PSM
Model	(1)	(2)	(3)
BigN	0.418***	0.260***	0.404***
	(20.29)	(4.16)	(17.68)
Size	0.420***	0.450***	0.428***
	(78.96)	(25.71)	(43.44)
Leverage	-0.053***	-0.032	-0.040
	(-3.77)	(-0.31)	(-1.53)
Loss	0.178***	0.168***	0.160***
	(12.47)	(4.32)	(7.25)
ROA	0.030***	0.054***	0.036***
	(4.86)	(2.68)	(2.92)
Cur_Ratio	0.009	-0.089	0.080
	(0.24)	(-0.85)	(1.44)
Quick_Ratio	-0.049***	-0.050***	-0.033***
	(-17.62)	(-8.80)	(-7.85)
ForSales	0.235***	0.242***	0.163***
	(18.54)	(7.36)	(8.06)
ln(BusSeg)	0.193***	0.235***	0.160***
	(11.17)	(3.83)	(5.70)
DecFYE	0.091***	0.086	0.030
	(5.63)	(1.60)	(1.26)
Going_Concern	0.015	0.040	0.015
	(0.57)	(0.77)	(0.46)
Fixed Effects	Industry, Year	Industry, Year	Industry, Year
Observations	16,884	16,884	5,558
R^2	0.761	0.711	0.574

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Table 4, continued

Panel B: Difference-in-differences estimate of the Big-N fee premium over the 2000 – 2004 period

Specification	OLS	EB	PSM
Model	(1)	(2)	(3)
BigN	0.040	0.078	0.265***
	(1.57)	(0.95)	(9.26)
PostAA	0.091***	0.384***	0.151***
	(3.72)	(6.16)	(5.36)
BigN x PostAA	0.546***	0.238***	0.211***
	(20.39)	(3.98)	(5.68)
Size	0.424***	0.448***	0.431***
	(80.70)	(24.02)	(44.17)
Leverage	-0.055***	-0.034	-0.048*
	(-3.99)	(-0.31)	(-1.84)
Loss	0.176***	0.157***	0.161***
	(12.45)	(4.27)	(7.30)
ROA	0.025***	0.046**	0.034***
	(4.32)	(2.33)	(2.80)
Cur_Ratio	0.043	-0.095	0.089
	(1.12)	(-0.87)	(1.60)
Quick_Ratio	-0.049***	-0.047***	-0.032***
	(-17.98)	(-7.95)	(-7.91)
ForSales	0.233***	0.249***	0.165***
	(18.61)	(7.57)	(8.15)
ln(BusSeg)	0.194***	0.227***	0.162***
	(11.32)	(3.90)	(5.77)
DecFYE	0.093***	0.106*	0.034
	(5.81)	(1.92)	(1.47)
Going_Concern	0.046*	0.043	0.024
	(1.76)	(0.80)	(0.73)
Fixed Effects	Industry	Industry	Industry
Observations	16,884	16,884	5,558
R^2	0.754	0.692	0.567

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Table 5
Covariate balance statistics before and after entropy balancing, using approaches designed to minimize extreme observational weights

This table presents summary statistics and covariate distributions after modifying the baseline entropy balancing program (see Table 2) using two approaches designed to minimize the influence of control observations that are assigned large observational weights. In Panel A, we use entropy balancing to assign weights to the treatment sample of observations with a Big-N auditor (*BigN* = 1) rather than to the control sample of observations without a Big-N auditor. In Panel B, we exclude firms with accelerated filer status in 2004 from the samples of treatment and control observations, to focus on a sample of smaller firms with delayed implementation of Section 404 of the Sarbanes-Oxley Act. In both panels, we present covariate balance statistics before and after assigning weights to the sample of interest. Standardized differences (*Std Diff*) are presented to assess covariate balance on the first moment (means) between treatment and control samples, computed as the difference in means between the treatment and control samples divided by the square root of the average variance in the treatment and control samples for each covariate. The standardized difference will approach zero when the first moment of the distribution for a particular covariate is balanced across weighted and unweighted samples. Variance ratios are presented to assess covariate balance on the second moment (variance), computed as the ratio of the variance of each covariate in the unweighted sample (firms without a Big-N auditor in Panel A, non-accelerated filers with a Big-N auditor in Panel B) scaled by variance for the weighted sample. The *Match Ratio* and *Maximum Weight* are calculated using weights identified for the control sample of observations using entropy balancing to achieve balance on the first and second distributional moments (see McMullin and Schonberger 2020). Refer to Appendix A for detailed variable definitions and data sources.

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Table 5, continued

Panel A: Covariate balance statistics before and after using entropy balancing to assign weights to firms with a Big-N auditor (treatment) rather than to non-Big-N (control) observations

Prior to balancing:	**Firms without a Big-N auditor (3,969 Observations)**			**Firms with a Big-N auditor (12,915 Observations)**			**Balance Stats**	
Covariate	**Mean**	**Variance**	**Skewness**	**Mean**	**Variance**	**Skewness**	**Std Diff**	**Variance Ratio**
Size	3.423	2.245	0.369	6.267	3.960	0.209	-1.615 [a]	0.567 [a]
Leverage	0.515	1.155	4.079	0.229	0.091	6.711	0.363 [a]	12.699 [a]
Loss	0.620	0.236	-0.494	0.409	0.242	0.372	0.432 [a]	0.975
ROA	-0.901	7.205	-4.616	-0.118	0.410	-14.170	-0.401 [a]	17.582 [a]
Cur_Ratio	0.562	0.073	-0.221	0.510	0.063	-0.029	0.198 [a]	1.161
Quick_Ratio	1.965	10.360	4.758	2.554	11.530	4.117	-0.178 [a]	0.899
ForSales	0.246	0.341	2.866	0.529	0.544	1.459	-0.426 [a]	0.626 [a]
ln(BusSeg)	0.894	0.143	1.023	0.981	0.244	0.282	-0.197 [a]	0.587 [a]
DecFYE	0.621	0.235	-0.499	0.697	0.211	-0.858	-0.161 [a]	1.115
Going_Concern	0.274	0.199	1.014	0.048	0.046	4.205	0.644 [a]	4.313 [a]
After balancing:								
Covariate	**Mean**	**Variance**	**Skewness**	**Mean**	**Variance**	**Skewness**	**Std Diff**	**Variance Ratio**
Size	3.423	2.245	0.369	3.423	2.245	0.317	0.000	1.000
Leverage	0.515	1.155	4.079	0.515	1.155	3.729	0.000	1.000
Loss	0.620	0.236	-0.494	0.620	0.236	-0.494	0.000	1.000
ROA	-0.901	7.205	-4.616	-0.901	7.205	-4.727	0.000	1.000
Cur_Ratio	0.562	0.073	-0.221	0.562	0.073	-0.239	0.000	1.000
Quick_Ratio	1.965	10.360	4.758	1.965	10.360	5.462	0.000	1.000
ForSales	0.246	0.341	2.866	0.246	0.341	2.922	0.000	1.000
ln(BusSeg)	0.894	0.143	1.023	0.894	0.143	0.799	0.000	1.000
DecFYE	0.621	0.235	-0.499	0.621	0.235	-0.499	0.000	1.000
Going_Concern	0.274	0.199	1.014	0.274	0.199	1.014	0.000	1.000
Match Ratio	**0.22**	Ratio: Firms with above equal weights in e-balance / All control firms						
Maximum Weight	**53.41**	Maximum weight assigned to a single control sample observation by e-balance						
After trimming one Big-N observation with the largest observational weight:								
Match Ratio	**0.21**	Ratio: Firms with above equal weights in e-balance / All control firms						
Maximum Weight	**30.93**	Maximum weight assigned to a single control sample observation by e-balance						

[a] Indicates covariates with standardized differences (variance ratios) outside of the +/- 0.1 (4/5 and 5/4) bounds suggested by Rubin (2001) as indicating a balanced covariate.

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Table 5, continued

Panel B: Covariate balance statistics before and after using entropy balancing on the subsample of non-accelerated filers to assign weights to firms without a Big-N auditor (control sample)

Prior to balancing:	**Firms with a Big-N auditor (2,992 Observations)**			**Firms without a Big-N auditor (2,905 Observations)**			**Balance Stats**	
Covariate	**Mean**	**Variance**	**Skewness**	**Mean**	**Variance**	**Skewness**	**Std Diff**	**Variance Ratio**
Size	4.558	2.456	0.841	3.122	1.534	-0.051	1.017 [a]	1.601 [a]
Leverage	0.249	0.162	7.078	0.563	1.318	3.786	-0.365 [a]	0.123 [a]
Loss	0.554	0.247	-0.215	0.649	0.228	-0.625	-0.196 [a]	1.085
ROA	-0.247	0.861	-10.610	-1.003	8.034	-4.347	0.359 [a]	0.107 [a]
Cur_Ratio	0.559	0.068	-0.233	0.568	0.074	-0.244	-0.032	0.915
Quick_Ratio	2.573	12.730	4.015	1.919	11.040	4.891	0.190 [a]	1.153
ForSales	0.437	0.520	1.850	0.227	0.335	3.055	0.321 [a]	1.555 [a]
ln(BusSeg)	0.933	0.179	0.550	0.875	0.132	1.173	0.146 [a]	1.357 [a]
DecFYE	0.667	0.222	-0.709	0.613	0.237	-0.464	0.113 [a]	0.936
Going_Concern	0.100	0.090	2.674	0.307	0.213	0.835	-0.534 [a]	0.421 [a]
After balancing:								
Covariate	**Mean**	**Variance**	**Skewness**	**Mean**	**Variance**	**Skewness**	**Std Diff**	**Variance Ratio**
Size	4.558	2.456	0.841	4.558	2.456	0.333	0.000	1.000
Leverage	0.249	0.162	7.078	0.249	0.163	7.262	0.000	0.997
Loss	0.554	0.247	-0.215	0.554	0.247	-0.215	0.000	1.000
ROA	-0.247	0.861	-10.610	-0.247	0.865	-8.745	0.000	0.995
Cur_Ratio	0.559	0.068	-0.233	0.559	0.068	-0.203	0.000	1.000
Quick_Ratio	2.573	12.730	4.015	2.573	12.730	3.930	0.000	1.000
ForSales	0.437	0.520	1.850	0.437	0.520	1.725	0.000	1.000
ln(BusSeg)	0.933	0.179	0.550	0.933	0.179	0.696	0.000	1.001
DecFYE	0.667	0.222	-0.709	0.667	0.222	-0.709	0.000	1.000
Going_Concern	0.100	0.090	2.674	0.100	0.090	2.674	0.000	1.000
Match Ratio	**0.27**	Ratio: Firms with above equal weights in e-balance / All control firms						
Maximum Weight	**72.45**	Maximum weight assigned to a single control sample observation by e-balance						
After trimming one non-Big-N observation with the largest observational weight:								
Match Ratio	**0.24**	Ratio: Firms with above equal weights in e-balance / All control firms						
Maximum Weight	**65.26**	Maximum weight assigned to a single control sample observation by e-balance						

[a] Indicates covariates with standardized differences (variance ratios) outside of the +/- 0.1 (4/5 and 5/4) bounds suggested by Rubin (2001) as indicating a balanced covariate.

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Table 6
Regression estimates of the Big-N audit fee premium for the full sample of observations, after implementing strategies to reduce the influence of extreme observational weights

This table presents results of entropy-balance- (EB-) weighted and propensity-score-matched (PSM) regression estimates of Eq. (7) over the 2000 – 2004 sample window (with 2002 omitted as a transition year) for the full sample of observations, where the dependent variable is the natural logarithm of annual audit fees, $ln(Fee_t)$. In models (1) and (2), we use EB to assign weights to the treatment sample of observations with a Big-N audit firm (*BigN* = 1) using the full sample of observations, resulting in an estimate of the average treatment effect for the control observations (*ATC*). Model (2) uses a subsample of observations remaining after deleting one observation with the largest weight assigned by EB to the treatment sample of observations (*BigN* = 1), as described in Panel A of Table 5. In model (3), we present results for propensity-score-matched (*PSM*) regression estimates after trimming observations with extreme propensity scores below 0.05 and above 0.95. Refer to Section 4 for details of the PSM program and to Table 3 for Logit model estimates of the propensity score. In model (4), we estimate EB-weighted regressions with weights assigned to the control sample of observations without a Big-N audit firm (*BigN* = 0) using a subsample of observations remaining after trimming observations with extreme propensity scores below 0.05 and above 0.95 from the full sample (*TRIM – EB*). The independent variables of interest are an indicator variable set equal to one for firm-year observations with a Big-N auditor (*BigN*=1) and to zero otherwise, an indicator variable set equal to one for fiscal years 2003 – 2004 following the August 2002 demise of Arthur Andersen (*PostAA* = 1) and to zero otherwise, and the interaction between these variables (*BigN* × *PostAA*) capturing our difference-in-differences estimate of the Big-N audit fee premium. Refer to Appendix A for detailed variable definitions and data sources. All models include control variables appearing in Eq. (7) as well as 2-digit SIC industry-fixed effects. t-statistics computed using standard errors clustered by firm appear in parentheses below each coefficient. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Specification	**EB (ATC estimate)**		**PSM**	**TRIM – EB**
Obs trimmed?	No	Yes	Yes	Yes
Model	(1)	(2)	(3)	(4)
BigN	0.302***	0.307***	0.279***	0.176***
	(11.15)	(11.16)	(9.72)	(4.44)
PostAA	0.102***	0.102***	0.145***	0.343***
	(4.17)	(4.17)	(5.10)	(7.48)
BigN x PostAA	0.190***	0.179***	0.182***	0.168***
	(4.51)	(4.15)	(4.89)	(3.43)
Additional control variables included?	Yes	Yes	Yes	Yes
Fixed Effects	Industry	Industry	Industry	Industry
Observations	16,884	16,883	5,241	11,099
R^2	0.531	0.529	0.425	0.484

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Table 7

Regression estimates of the Big-N audit fee premium for the subsample of non-accelerated filers surrounding the demise of Arthur Andersen

This table presents regression estimates of Eq. (7) using the subsample of non-accelerated filers over the 2000 – 2004 sample window (with 2002 omitted as a transition year), where the dependent variable is the natural logarithm of annual audit fees, $ln(Fee_t)$. The independent variables of interest are an indicator variable set equal to one for firm-year observations with a Big-N auditor (*BigN*=1) and to zero otherwise, an indicator variable set equal to one for fiscal years 2003 – 2004 following the August 2002 demise of Arthur Andersen (*PostAA* = 1) and to zero otherwise, and the interaction between these variables (*BigN* × *PostAA*) capturing our difference-in-differences estimate of the Big-N audit fee premium. Model (1) presents an ordinary least squares (*OLS*) regression estimate of the audit fee model. In models (2) and (3), we use entropy balancing (*EB*) over the subsample of non-accelerated filers to assign weights to the control sample of observations without a Big-N audit firm (*BigN* = 0). Models (4) and (5) present propensity-score-matched (*PSM*) regression estimates using all observations with non-accelerated filer status with a valid match, without replacement (in model 4) or after trimming observations with extreme propensity scores less than 0.05 or more than 0.95 (in model 5). In model (6), we trim observations with extreme propensity scores less than 0.05 or more than 0.95 from the sample of non-accelerated filers before estimating EB-weighted regressions on the remaining sample of observations (*TRIM – EB*). Refer to Section 4 for details of the PSM program and to Table 3 for Logit model estimates of the propensity score. Refer to Appendix A for detailed variable definitions and data sources. All regression models include control variables appearing in Eq. (7) as well as 2-digit SIC industry-fixed effects. t-statistics computed using standard errors clustered by firm appear in parentheses below each coefficient. *, **, and *** indicate significance at the 10%, 5%, and 1% two-tailed levels, respectively.

Specification	**OLS**	**EB**		**PSM**		**TRIM – EB**
Obs trimmed?	*No*	*No*	*Yes*	*No*	*Yes*	*Yes*
Model	(1)	(2)	(3)	(4)	(5)	(6)
BigN	0.229***	0.352***	0.375***	0.285***	0.284***	0.330***
	(6.75)	(5.12)	(4.99)	(7.46)	(7.44)	(5.59)
PostAA	0.048	0.217***	0.238***	0.095***	0.098***	0.195***
	(1.64)	(3.12)	(3.13)	(2.71)	(2.81)	(3.31)
BigN x PostAA	0.319***	0.152**	0.133*	0.210***	0.213***	0.184***
	(8.57)	(2.16)	(1.75)	(4.79)	(4.87)	(3.02)
Additional control variables included?	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects	Industry	Industry	Industry	Industry	Industry	Industry
Observations	5,897	5,897	5,896	3,614	3,576	5,540
R^2	0.632	0.569	0.576	0.431	0.414	0.463

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Table 8

Covariate balance statistics before and after entropy balancing in individual annual periods for observations with a Big-N auditor (treatment) and observations without a Big-N auditor (control)

This table presents summary statistics and covariate distributions after using entropy balancing to assign weights to observations without a Big-N audit firm (*BigN* = 0) using two approaches to adjust weights for the panel dataset. Panel A presents covariate balance statistics after assigning weights to observations in only the last fiscal year (2001) prior to the demise of Arthur Andersen. These annual observational weights are then assigned to remaining years in the panel (2000, 2003, and 2004), thereby restricting the sample to firm-year observations with data available to estimate the audit fee model in Eq. (1) in the 2001 fiscal year. Panel B presents covariate balance statistics after assigning weights to observations separately in each fiscal year over the 2000 – 2004 sample window. Standardized differences (*Std Diff*) are presented to assess covariate balance on the first moment (means) between treatment and control samples, computed as the difference in means between the treatment and control samples divided by the square root of the average variance in the treatment and control samples for each covariate. The standardized difference will approach zero when the first moment of the distribution for a particular covariate is balanced across weighted and unweighted samples. Variance ratios are presented to assess covariate balance on the second moment (variance), computed as the ratio of the variance of each covariate in the treatment sample (firms with a Big-N auditor) scaled by variance for the weighted control sample (firms without a Big-N auditor). The *Match Ratio* and *Maximum Weight* are calculated using weights identified for the control sample of observations using entropy balancing to achieve balance on the first and second distributional moments (see McMullin and Schonberger 2020). Refer to Appendix A for detailed variable definitions and data sources.

Panel A: Covariate distributions after using entropy balancing in the 2001 fiscal year only for firms with a Big-N auditor (treated) vs. firms without a Big-N auditor (control)

	Firms with a Big-N auditor (3,512 Observations)			Firms without a Big-N auditor (706 Observations)			Balance Stats	
Covariate	**Mean**	**Variance**	**Skewness**	**Mean**	**Variance**	**Skewness**	**Std Diff**	**Variance Ratio**
Size	6.267	3.960	0.209	6.090	4.384	0.076	0.087	0.903
Leverage	0.229	0.091	6.711	0.230	0.138	8.152	-0.005	0.659 [a]
Loss	0.409	0.242	0.372	0.435	0.246	0.264	-0.052	0.983
ROA	-0.118	0.410	-14.170	-0.155	0.732	-12.680	0.049	0.560 [a]
Cur_Ratio	0.510	0.063	-0.029	0.532	0.059	-0.281	-0.087	1.063
Quick_Ratio	2.554	11.530	4.117	2.495	9.320	4.666	0.018	1.237
ForSales	0.529	0.544	1.459	0.445	0.407	1.684	0.122 [a]	1.338 [a]
ln(BusSeg)	0.981	0.244	0.282	0.989	0.241	0.126	-0.016	1.011
DecFYE	0.697	0.211	-0.858	0.669	0.222	-0.716	0.061	0.953
Going_Concern	0.048	0.046	4.205	0.063	0.059	3.595	-0.064	0.780 [a]
Match Ratio	**0.46**	Ratio: Firms with above equal weights in e-balance / All control firms						
Maximum Weight	**272.99**	Maximum weight assigned to a single control sample observation by e-balance						

[a] Indicates covariates with standardized differences (variance ratios) outside of the +/- 0.1 (4/5 and 5/4) bounds suggested by Rubin (2001) as indicating a balanced covariate.

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Table 8, continued

Panel B: Covariate distributions after using entropy balancing in each fiscal year over the 2000 – 2004 sample window for firms with a Big-N auditor (treated) vs. firms without a Big-N auditor (control)

	Firms with a Big-N auditor (12,915 Observations)			Firms without a Big-N auditor (3,969 Observations)			Balance Stats	
Covariate	**Mean**	**Variance**	**Skewness**	**Mean**	**Variance**	**Skewness**	**Std Diff**	**Variance Ratio**
Size	6.267	3.960	0.209	6.267	3.957	0.000	0.000	1.001
Leverage	0.229	0.091	6.711	0.229	0.092	4.875	-0.001	0.993
Loss	0.409	0.242	0.372	0.409	0.242	0.372	0.000	1.000
ROA	-0.118	0.410	-14.170	-0.118	0.415	-11.060	0.001	0.988
Cur_Ratio	0.510	0.063	-0.029	0.510	0.063	-0.140	0.000	1.001
Quick_Ratio	2.554	11.530	4.117	2.554	11.520	4.445	0.000	1.001
ForSales	0.529	0.544	1.459	0.529	0.544	1.504	0.000	1.001
ln(BusSeg)	0.981	0.244	0.282	0.981	0.244	0.203	0.000	1.001
DecFYE	0.697	0.211	-0.858	0.697	0.211	-0.858	0.000	1.000
Going_Concern	0.048	0.046	4.205	0.049	0.046	4.203	0.000	0.999
Match Ratio	**0.17**	Ratio: Firms with above equal weights in e-balance / All control firms						
Maximum Weight	**282.56**	Maximum weight assigned to a single control sample observation by e-balance						

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Table 9

Matched sample estimates of the Big-N audit fee premium, where matching is conducted in individual years of the panel dataset

This table presents results of entropy-balance-weighted (*EB*) and propensity-score-matched (*PSM*) regression estimates of Eq. (7) using firm-year observations over the 2000 – 2004 sample window (with 2002 omitted as a transition year), where the dependent variable is the natural logarithm of annual audit fees, $ln(Fee_t)$. In model (1), we use EB to assign weights to the control sample of observations without a Big-N auditor (*BigN* = 0) using covariate values observed in the last fiscal year (*2001 only*) prior to the demise of Arthur Andersen. These annual observational weights are then assigned to remaining years in the panel (2000, 2003, and 2004), thereby restricting the sample to firm-year observations with data available to estimate the audit fee model in the 2001 fiscal year. Similarly, in model (2), we use PSM to match observations, without replacement, in the last fiscal year (*2001 only*) prior to the demise of Arthur Andersen, and retain this sample of matched observations in remaining years in the panel (2000, 2003, and 2004). The PSM program imposes a common support, uses a caliper of 0.03, and trims observations with extreme propensity scores less than 0.05 or more than 0.95. In model (3) ([4]), we use EB (PSM) to assign weights to the control sample of observations without a Big-N auditor (*BigN* = 0) separately in each fiscal year (*Separately each year*) over the 2000 – 2004 sample window (with 2002 omitted as a transition year). The sample underlying estimation of all models includes all firm-year observations in the full sample with data available for estimation. Independent variables of interest are an indicator variable set equal to one for firm-year observations with a Big-N auditor (*BigN*=1) and to zero otherwise, an indicator variable set equal to one for fiscal years 2003 – 2004 following the August 2002 demise of Arthur Andersen (*PostAA* = 1) and to zero otherwise, and the interaction between these variables (*BigN* × *PostAA*) capturing our difference-in-differences estimate of the Big-N audit fee premium. Refer to Table 6 for covariate balance and observational weighting statistics for EB models relying on the full sample. Refer to Section 4 for details of the PSM program and to Table 3 for Logit model estimates of the propensity score. Refer to Appendix A for detailed variable definitions and data sources. All models include control variables in Eq. (7) and 2-digit SIC industry-fixed effects. t-statistics computed using standard errors clustered by firm appear in parentheses below each coefficient. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Match year	**2001 only**		**Separately each year**	
Matching approach	*EB*	*PSM*	*EB*	*PSM*
Model	(1)	(2)	(3)	(4)
BigN	0.081	0.266***	0.109*	0.273***
	(1.29)	(8.24)	(1.73)	(9.11)
PostAA	0.500***	0.242***	0.432***	0.132***
	(9.59)	(8.94)	(7.61)	(4.40)
BigN x PostAA	0.152***	0.073*	0.196***	0.238***
	(2.86)	(1.91)	(3.56)	(6.20)
Additional controls?	Yes	Yes	Yes	Yes
Fixed Effects	Industry	Industry	Industry	Industry
Observations	14,229	3,359	16,884	4,552
R^2	0.739	0.466	0.714	0.515

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Table 10
Summary of the advantages and disadvantages of entropy balancing as a sample reweighting tool and implementation recommendations

This table presents a summary of the primary advantages and disadvantages when using entropy balancing (EB) as a tool to ensure covariate balance across treatment and control samples (Panel A) and summarizes our implementation recommendations for future researchers designed to address the primary disadvantages of EB in empirical research settings with various features (Panel B).

Panel A: Primary advantages and disadvantages of entropy balancing (EB)

Advantages of EB	***Disadvantages of EB***
Quickly achieves (near) exact covariate balance: EB ensures covariate balance (on multiple moments) without the need to specify a propensity score model, thus improving the tenability of the strong ignorability assumption of the Rubin Causal Model. Specifying a propensity score model often requires an iterative process of estimating the propensity score, matching, and balance checking to identify the model that maximizes covariate balance. Successful application of this iterative process proves difficult as improving balance on one covariate often impairs balance on another (Ho et al. 2007; Stuart et al. 2011; Hainmueller 2012).	*EB fails to converge when estimating weights:* Balance constraints may be too restrictive such that there does not an exist a set of positive weights that satisfy the balance constraints. Factors that contribute to a convergence issue include a higher number of balance conditions (more covariates included in the balancing algorithm or balancing on more distributional moment), a lack of distributional overlap between treatment and control samples in at least one covariate, or both.
Retains control sample observations: EB retains all control sample observations by allowing for continuous observational weights, rather than dropping (unmatched) observations. This approach maximizes estimation power and retains information in control sample (Watson and Elliot 2016).	*Assignment of extreme observational weights*: EB does not provide a simple way to exclude "extreme" treatment sample observations (i.e., those with a propensity score approaching either 0 or 1). Eliminating observations that do not possess a valid counterfactual in either the treatment or control sample requires specifying a propensity model (used to trim observations) or researcher-identification of a sample partition with greater distributional overlap.
Doubly robust estimation approach: EB is "doubly robust", such that if either the covariate balance conditions or the outcome regression model is correctly specified, the mean causal effect estimator is statistically consistent (Bang and Robins 2005; Zhao and Percival 2016).	*Incorporating panel data*: In settings with panel data involving multiple treatment "events", EB requires researchers to specify separate treatment and control samples for each event used to estimate control sample weights (see Goodman-Bacon 2021 for a discussion of this point). Note that all matching approaches display this requirement.
Computational cost: Hainmueller 2012 explains that EB is computationally attractive, with the optimization problem well-behaved and performed quickly even for large datasets.	*Requires a dichotomous treatment variable:* EB requires specifying a dichotomous treatment for its basic application. See Tübbicke (2020) for an extension of EB to continuous treatment variables.

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Post EB analysis: Hainmueller (2012) explains that resulting observational weights identified by EB are easily used in existing estimation packages (e.g., xtreg in Stata).	*Unobservables*: EB, as with all analysis in observational studies, does not ensure a match on unobserved (or omitted) covariates.

Panel B: Implementation recommendations for researchers when using entropy balancing (EB)

Research setting feature	***Recommendation***
Treatment and control samples display limited distributional overlap in at least one covariate	Estimate and present the maximum observational weight. In cases where this weight is large (relative to the number of control sample observations), consider the following steps before re-running EB: a) Restrict the treatment and control samples to a segment of the covariate distribution with more substantial overlap. This approach to identifying a local average treatment effect may utilize an observable characteristic (e.g., filing status) or a researcher identified threshold (e.g., firms with a market capitalization less than $1 billion) to partition observations. b) Estimate a propensity score model to identify observations with extreme propensity scores (near 0 or 1) or use an alternative matching algorithm (e.g., coarsened exact matching) to trim observations lacking a valid counterfactual in either the treatment or control samples.
Pool of treatment observations is substantially larger or more dispersed than the sample of control observations.	Consider estimating an average treatment effect on the control (ATC) by using EB to assign weights to treatment rather than control sample observations. This approach exploits the wider distribution of the treatment sample to select a more efficient match for each control sample observation.
Treatment of interest involves a panel dataset with multiple pre- and post-treatment periods	To avoid systematically over-weighting observations in either the pre- or post-period, use EB to assign weights in either: a) A single pre-period. This approach requires the availability of observations in the specified pre-period and assumes constant weights over the post-treatment-period. b) In each period of the sample. This approach allows both observations and weights to vary across time. However, this approach will tend to "balance away" variation related to the treatment of interest in the presence of covariates that are correlated with post-treatment variation.

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